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II Common Terms

A few words this manual uses throughout. The glossary (Section 22) explains the rarer ones.

`AT` command
`AT` stands for “attention”. An instruction to the modem, sent as text that starts with the letters `AT`, such as `ATZ`. One line can hold several commands (`ATE1Q0V1`).

Basic and extended commands
Basic commands are a letter, maybe after `&`, `\`, or `%`, and a number (`E1`, `&K3`). Extended commands start with `+` or `-` and accept values (`+MS=V92`).

Baud
Symbols per second: how many times a second the modem changes its signal. Each symbol can carry several bits, so a connection’s bit/s is often higher than its baud: `V.34` sends up to 3429 baud but 33600 bit/s. Older modems were sold by baud, which is why the two words are often mixed up.

bit/s
Bits per second, the speed of a connection; `CONNECT` gives it.

Carrier
The signal the far modem sends during a call. Losing it ends the call (`NO CARRIER`).

Command mode and online mode
In command mode the modem reads what the `DTE` sends as commands. Once a call is up (`CONNECT`) it is online: what the `DTE` sends goes to the far end as data. The escape sequence (`+++`, with a pause before and after) returns to command mode with the call kept, and `ATO` goes back online (Section 19).

DCE
Data Circuit-terminating Equipment. The modem: what sits between the `DTE` and the telephone line or call, here D-Modem itself.

Default
A setting’s value after `ATZ` or when the program starts. The modem keeps no stored profiles.

Dial string
What follows `ATD`: the number and the modifiers that change how it is dialed (`T`, `P`, `W`, `,` and so on).

Dry pair
A pair of copper wires with nothing on it: no telephone exchange, no battery voltage and no dial tone, such as two modems wired to each other directly. A modem uses it as a leased line; `V.wb` runs only on one.

DTE
Data Terminal Equipment. The computer or program that talks to the modem: a terminal program, a fax program, `pppd`, or the program that embeds libdmodem.

DTMF
Dual Tone Multi-Frequency. The tones of a telephone keypad, two at once for each key. The modem dials with them (`T`), and hears them in voice mode and in `DTMF` caller ID.

Error correction and compression
Error correction (`V.42`, `MNP`) resends data the line damaged; compression (`V.42bis`, `V.44`) packs it to go faster. Both need the far modem to have them too.

Fax
Facsimile. Sending a page as an image over a telephone line: the fax machine scans the page and its modem sends it to another fax machine, which prints it. This modem can act as a fax machine’s modem for a fax program on the `DTE` (Section 14).

ITU
International Telecommunication Union. The United Nations agency for telecommunications. Its standards sector, ITU-T, writes the recommendations modems follow: the V.... series for modems (`V.34`, `V.92`), T.... for fax (`T.30`), G.... for audio (`G.711`) and so on.

LAPM
Link Access Procedure for Modems. `V.42`’s error correction, the one most modems use: data goes in numbered frames with a checksum, and a damaged frame is sent again.

Leased line
A permanent connection between two modems, with no dialing, ringing or dial tone: each end starts at once (`&L`). A 2-wire leased line carries both directions on one pair; a 4-wire line has a pair for each direction.

MNP
Microcom Networking Protocol. An older family of error correction (classes 2 to 4) and compression (class 5), from before `V.42`; many modems still have it (`\N`).

Modem
Modulator-demodulator. A device that turns a computer’s data into sound that can travel over a telephone line, and turns the sound from the far end back into data. D-Modem does this in software, with no modem hardware.

Modulation
How the modem turns data into sound on the line, and so how fast it goes: `V.34`, `V.90`, `V.92` and the rest, chosen with `+MS`.

PCM
Pulse Code Modulation. Sound as numbers: the telephone network carries a call

---

Table 1. AT command summary *(continued)*

Command	Description	Command	Description
`&Fn`	Factory *defaults* (accepted).	`+IFC, +ICF, +IPR`	`DTE` interface.
`&Gn`	Guard tone (accepted).	`+ILRR, +IDSR, +ILSD`	`DTE` interface (kept).
`&Hn`	Flow control (kept).	`+MR, +ER, +DR`	Reports (kept).
`&In`	`XON`/`XOFF` flow control (accepted).	`+PCW, +PIG`	`V.92` options (kept).
`&Jn`	Auxiliary relay (accepted).	`+PMH, +PMHT`	Modem-on-hold.
`&Kn`	Data compression.	`+PMHR`	Ask for modem-on-hold.
`&Ln`	Leased line.	`+CVM`	`V.cvm`.
`&Mn`	Asynchronous mode (accepted).	`-SEC`	`MNP 10EC` (kept).
`&Nn`	Line rate (accepted).	`+FCLASS`	Data, fax or voice.
`&Pn`	Pulse dial ratio (kept).	`+FAA`	Adaptive answer.
`&Qn`	Error control mode.	`+FMI, +FMM, +FMR`	Fax identification.
`&Rn`	`RTS`/`CTS` (accepted).	`+FTS, +FRS`	Fax silence.
`&Sn`	Data set ready (accepted).	`+FTM, +FRM`	Fax data.
`&Tn`	Loop tests.	`+FTH, +FRH`	Fax `HDLC` frames.
`&Vn`	View active configuration, call statistics.	`+FPR`	Fax `DTE` rate (kept).
`&Wn`	Store profile (accepted).	`+V...`	Voice (Section 18).
`&Yn`	Profile at power on (accepted).	`+FLI, +FPI`	Fax `Class 2` identity.
		`+FLP, +FSP`	Fax `Class 2` polling.
		`+FCC, +FIS, +FCS`	Fax `Class 2` capabilities.
		`+FBO, +FHS, ...`	Fax `Class 2` settings.
		`+FDT, +FDR, +FET, +FKS`	Fax `Class 2` pages.
		`+MV18S, +MV18R`	`V.18` text telephone.
		`+MV18AM, +MV18P`	`V.18` answer message.
		`+MBAUDOT`	5-bit (`TTY`, telex) settings.
		`+A8E`	`V.8` and `V.8 bis`.
		`+DSVD`	`V.70` voice and data (`DSVD`).

I Introduction

This manual specifies the `AT` command set of D-Modem's modem core (`core/` in the source). The core accepts the commands of every D-Modem program:

- **`dmodemd`**, the modem on a pseudo terminal (`/dev/ttyDM0` links to it), its line a `SIP` call or, with `-a` and `-usb`, an analogue telephone line;
- **libdmodem** (`dmodem.h`), the modem as a library: the program that embeds it sends the commands;
- the **Asterisk and FreeSWITCH modules** (`app_dmodem`, `mod_dmodem`), whose profiles give the commands each call starts with.

Feature Overview

- **Data** from 300 bit/s to 56000 bit/s: `V.21`, `V.22`, `V.22bis`, `V.23`, Bell 103 and 212A, `V.32`, `V.32bis`, `V.34` up to 33600 bit/s, and `V.90` and `V.92` as the digital modem. Automode finds the best modulation both ends have (Section 7).
- **Error correction and compression**: `V.42` (`LAPM`) and `MNP` 2 to 5; `V.42bis`, `V.44`, `MNP 5` and zstd (Sections 8 and 9).
- **V.92's extras**: quick connect, `PCM` upstream and modem-on-hold, to answer a waiting call and come back to the data call.
- **Fax**: `Class 1` (`T.31`) and `Class 2` and `Class 2.0` (`T.32`), up to 14400 bit/s with `V.17`, and adaptive answer, which tells a fax call from a data call (Section 14).
- **Voice** (`V.253`): play and record audio, send and detect `DTMF` and tones, use a handset, and turn a voice call into a fax or data call (Section 18).
- **Text telephones** (`V.18`): the 5-bit `TTY` of the deaf community, telex and `RTTY`, `EDT`, and automoding between them (Section 15).
- **Caller ID**, from a `SIP` call or a telephone line (`FSK` and `DTMF`), and a waiting call's caller in voice mode.
- **Leased lines**, 2-wire and 4-wire (`V.29`, `V.33`, `V.27ter`), with `V.13`'s simulated carrier control.
- **Digital channels**: `V.91` between two digital modems, the clear 64000 bit/s channel of an `ISDN` data call (`X.75`, `PPP`, `V.110`, `V.120`) (Section 11), and `V.80`'s synchronous access.
- **V.cvm** for data over mobile speech paths, and **V.wb**, a wideband modem for a dry pair.
- **Diagnostics**: the last call's statistics (`ATI11`, `AT&V1`, `AT#UD`) and loop tests (`AT&T`).
- **Other modems' setup strings** work: the modem accepts the commands of `V.250`, Rockwell and Conexant, and USRobotics modems, most of them with no effect; their entries say so.

Every command, range, *default* and response in this manual is the core's own. A setting that is kept but changes nothing says so.

How to Read This Manual

Each command has an entry of the same form: its name and title in a strip, what it does, then where it has them its syntax, its result codes and a numbered table of its values, the *default* marked *(default)*. Table 1 lists every command and links to its entry. Commands are shown in capitals; the modem accepts either case. *n* is a decimal number, *h* a hexadecimal one; `[ ]` marks what may be left out. A setting's *default* is the one `ATZ` and the start of the program give.

The text is styled by what it is:

- `Red monospace` is a command or part of one, as you type it (`ATD`, `+MS=V92`); where it is a link it leads to that command's entry.
- `Monospace on blue` is what the modem sends back: a result code or other response, as it appears on the `DTE`'s screen (`OK`, `CONNECT 33600`, `NO CARRIER`).
- `Plain monospace` is a standard, a signal or a term from one (`V.34`, `CNG`), an acronym (`DTE`), or a number such as a speed (33600 bit/s).
- *Italics* stand for a value you fill in (`ATS`*n*).

as 8000 8-bit samples a second (`G.711`). `V.90` and `V.92` send data as these numbers, which is how they reach 56000 bit/s.

Result code

The modem's answer to a command line, or news of a call: `OK`, `ERROR`, `CONNECT`, `RING`, `NO CARRIER` and so on (Section 12).

S-register

A numbered setting, read with `ATS`*n*`?` and set with `ATS`*n*`=`*value* (Section 13).

SIP

Session Initiation Protocol. The protocol of internet telephony. `dmodemd` places and answers its calls as `SIP` calls (Section 20).

TTY

Teletypewriter. A text telephone used by deaf and hard of hearing people, sending Baudot code at `45.45` or `50` bit/s (Section 15). On Linux, a tty is also a terminal device in `/dev`, such as a serial port (`/dev/ttyS0`) or a pseudo terminal (`/dev/pts/3`). `dmodemd` appears as a pseudo terminal, so terminal programs open it like a serial modem.

III Command Summary

Table 1. AT command summary

Command	Description	Command	Description
`A/`	Repeat last command.	`\A`*n*	`MNP` block size.
`A`	Answer.	`\G`*n*	`XON`/XOFF pass through (accepted).
`B`*n*	Communication standard.	`\J`*n*	Rate adjust (accepted).
`C`*n*	Carrier control (accepted).	`\K`*n*	Break control (accepted).
`D`*s*	Dial.	`\N`*n*	Error control mode.
`E`*n*	Command echo.	`\Q`*n*	Flow control (accepted).
`F`*n*	On-line data echo (accepted).	`\R`*n*	Ring indicator (accepted).
`H`*n*	Hook control.	`\T`*n*	Inactivity timer (accepted).
`I`*n*	Identification, call statistics.	`\V`*n*	Protocol in `CONNECT`.
`L`*n*	Speaker volume.	`%B, *B`	Blacklisted numbers.
`M`*n*	Speaker control.	`%C`*n*	Data compression.
`N`*n*	Automode.	`%E`*n*	Line quality monitor (kept).
`O`*n*	Return to on-line data mode.	`#CID`	Caller ID.
`P`	Pulse dialing.	`#UD`	Unimodem diagnostics.
`Q`*n*	Result codes on or off.	`-C, -K, -Q`	`MNP 10` (accepted).
`S`*n*	Read or set an S-register.	`)M, @M, *H`	`MNP 10` (accepted).
`T`	Tone dialing.	`+MS`	Modulation.
`V`*n*	Result code form.	`+GMI, +GMM, +GMR`	Identification.
`W`*n*	Connect message (accepted).	`+GCAP`	Capabilities.
`X`*n*	Result codes and call progress.	`+GCI`	Country.
`Y`*n*	Long space disconnect (kept).	`+ES`	Error control.
`Z`*n*	Reset.	`+ESA`	`V.80` synchronous access.
`&A`*n*	Connection lines before `CONNECT`.	`+ITF`	`V.80` transmit buffer.
`&B`*n*	`V.32` retrain (accepted).	`+DS`	`V.42bis`.
`&C`*n*	Data carrier detect (kept).	`+DS44`	`V.44`.
`&D`*n*	Data terminal ready (accepted).	`+DSZ`	zstd compression.
`&E`*n*	Rate in `CONNECT` (kept).	`+B2, +CBST, +CRC`	B-channel protocol, `V.110`, ring type.

3 Settings and Profiles

`S`*n* — Read or Set an S-Register

`ATS`*n*`?` reads S-register *n* (0 to 255) and gives its value in decimal on a line of its own. `ATS`*n*`=`*v* sets it to *v*; the registers hold 8 bits, so *v* is taken modulo 256. Both numbers are decimal, a leading 0 too (`ATS09?` is S9). `ATS103=`*n* is `\N`*n*. Section 13 lists the registers.

Syntax:

- `ATS`*n*`?`
- `ATS`*n*`=`*v*

Result code	When
`OK`	Always.
`ERROR`	For a number over 255, a read-only register (S44, S45), or neither `?` nor `=` after it.

`Z`*n* — Reset

This command hangs up and sets the modem as it was at the start: the S-registers, `+MS`, modem-on-hold, the data mode (`+FCLASS=0`), the extended parameters and the loop tests (`&T5`). The country stays, and so do the S-registers without a *default* of their own: S18, S28, S47 (`&L`), S48, S161 and the free ones. There are no stored profiles, so *n* (0 to 3) changes nothing. The rest of the line is not read.

Result code	When
`OK`	Always.

`&F`*n* — Restore Factory Defaults

Accepted, with or without a digit, and nothing done; see `&D`. `ATZ` restores the *defaults*.

Table 1. AT command summary *(continued)*

Command	Description	Command	Description
		`+ASVD`	`V.61` voice and data (`ASVD`).
		`+TMO`	`V.59` diagnostics.
		`+TRDL`, `+TAL`, `+TTER`, ...	Loop and error rate tests (`V.58`).

`V`*n* — Result Code Form

This command sets whether result codes are words or numbers (S15). Section 12 lists both.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 4. V*n* Commands

Command	Function
`V0`	Numbers, each on a line of its own.
`V1`	Words (*default*).

`W`*n* — Connect Message Control

Other modems use `W` to choose what `CONNECT` reports. The modem accepts it and changes nothing: here `&A` and `\V` choose.

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

`X`*n* — Result Code Set and Call Progress

This command sets whether `CONNECT` gives the rate, whether the modem waits for a dial tone before dialing, and whether it reports a busy tone (S56).

Result code	When
`OK`	If *n* = 0–4.
`ERROR`	Otherwise.

Table 5. X*n* Commands

	CONNECT	Before dialing	BUSY
`X0`	Without the rate.	No wait for a dial tone.	No.
`X1`	With the rate.	No wait for a dial tone.	No.
`X2`	With the rate.	Waits for a dial tone (below).	No.
`X3`	With the rate.	No wait for a dial tone.	Yes.
`X4`	With the rate.	Waits for a dial tone (below) (*default*).	Yes.

The dial tone `X2` and `X4` wait for is 1 s of tone, steady or in a cadence whose gaps are 1 s at most (as Italy's: its tone added up); `NO DIALTONE` if none comes in S6 seconds (2 s at least; 2 s more while a dial tone in a cadence is heard). A busy tone is four cycles of the country's busy or congestion tone's cadence (the `ITU`'s list's, 15% or 80 ms either side: 0.48 s on and off or 0.24 in Germany, 0.75 s in India, 1 s on and 4 off in Egypt), not its ringing tone; a country with neither (none of the 192) its table's bounds. `NO ANSWER` comes with every *n*, and the dial string's `W` and `@` wait with every *n*. A `SIP` call has no dial tone: `dmodemd` wants `ATX3`.

1 Command Lines

`AT` commands are issued to the modem to control its operation and its settings. The modem accepts them in command mode: after it starts, when a call ends, after any command but a dial or an answer, and online after the escape sequence (Section 19), the call still up. Each command line is answered with a result code (Section 12).

- A command line begins with `AT` and ends with the character of S3 (carriage return) or S4 (line feed). Blanks and tabs in it are dropped before it is read.
- The character of S5 (backspace) erases the last one typed.
- A line holds **63** characters at most; a longer one is `ERROR`.
- With echo on (`E1`), what the `DTE` types comes back as it is typed.
- A line that does not begin with `AT` is not a command: with echo on it is blanked out on the `DTE`'s screen.

The commands of a line run in turn (`ATE1Q0V1`). A basic command's number left out is 0 (`ATE` is `ATE0`). An extended command (`+...`) accepts `=`*values*, `?` (its values) and `=?` (its ranges). A `;` may stand between two commands, as `V.250` puts one after an extended command that another follows (`AT+IFC=2,2;+MS=V92`); here it is not needed. A value left out of an extended command's `=`*values* keeps what it was.

Some extended parameters (`+ILRR`, `+MR`, `+PCW`, `-SEC` and others) are kept for other modems' setup strings, with no other effect but where an entry says so. `=`*values* sets them (each in its range, else `ERROR`), `?` gives `+`*`name`*`: `*`values`*, `=?` the ranges, and `ATZ` the *defaults*.

The commands are grouped by what they do, each group a section: echo and result codes, settings and profiles, identification, dialling and answering, the speaker and the line, modulation, error control, data compression, the `DTE` interface, `ISDN`, then fax, text telephones and voice. Table 1 lists them by name.

The first command that fails ends the line with `ERROR`; otherwise `OK` follows the last one. `A`, `D` and `O` end the line and give their result later (`CONNECT`, `NO CARRIER` and so on), and `Z` ends it with `OK`: whatever follows them on the line is not read, but for a dial string's `;`, after which the rest of the line runs once the number is dialed (`ATD`). A command that reports something (`I`, `S`*n*`?`, `+MS?` . . .) prints it on lines of its own before the result.

A character from the `DTE` while a call is being set up (dialing, answering, the modems' training) hangs up. After a dial string's `;` the modem is in command state, and accepts commands.

---

2 Echo and Result Code Settings

`A/` — Repeat Last Command

`A/` runs the last command line again. It needs no `AT` before it and no line end after it: the modem runs it as the `/` is typed. It is mostly used to dial a busy number again.

`E`*n* — Command Echo

This command sets whether the modem echoes the `DTE`'s characters in command mode (S13).

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 2. E*n* Commands

Command	Function
`E0`	No echo.
`E1`	Echo (*default*).

`Q`*n* — Result Code Control

This command sets whether the modem gives result codes (S14).

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 3. Q*n* Commands

Command	Function
`Q0`	Result codes are given (*default*).
`Q1`	Result codes are not given.

`&A`*n* — Connection Lines Before CONNECT

This command adds lines on the connection before a data call's `CONNECT` (S70). *n* is a sum of the bits below. With `V0` the first three lines are numbers and the fourth is left out.

Result code	When
`OK`	If *n* = 0–7.
`ERROR`	Otherwise.

Table 6. &A*n* Bits

Bit	Line	With V0
`1`	`Modulation: `*`name`*, the modulation (V92, V34, V32bis, . . .).	1000 and its number (1092, 1034, 1132, . . .).
`2`	`Protocol: LAPM`, `MNP` or `NONE`, the error correction (`V.42`'s `LAPM`, `MNP` 2 to 4).	77, 78, 76.
`1 and 2`	`Compression: V44`, `V42bis`, `MNP5`, `ZSTD` (`V.cvm`'s, Annex F) or `NONE`, what `V.42` or `MNP` agreed.	68, 67, 66, 70, 69.
`4`	`TxRate: `*`rate`*, the rate the modem sends at.	None.

The *default* is 0, no lines.

`&E`*n* — Rate in CONNECT

The modem keeps the value in S71. It changes nothing: `CONNECT` always gives the rate the modem receives at.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

`\V`*n* — Protocol Result Code

This command sets whether a data call's `CONNECT` gives the error correction and the compression after its rate (S106). With `V0` or `X0` it changes nothing.

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

Table 7. \V*n* Commands

Command	Function
`\V0`	The rate alone: `CONNECT 33600` (*default*).
`\V1, \V2`	The protocol too: `CONNECT 33600/LAPM/V42bis`, `CONNECT 2400/MNP/MNP5`, `CONNECT 28800/NONE`.

`+MR, +ER, +DR` — Modulation, Error Control and Compression Reports

On other modems these turn on the `+MCR:`, `+MRR:`, `+ER:` and `+DR:` reports. Kept; each 0 or 1, *default* 0.

`&Vn` — View Active Configuration

`&V` and `&V0` show the active settings, in lines of **72** columns at most: the basic commands', the S-registers in use (`S`*nn*`:`*vvv*; not S18 or S44 to S48), +MS and the other extended settings, and the extended parameters. There are no stored profiles: the list ends with `STORED PROFILES: none`. `&V1` gives the last call's statistics, as `ATI11` (Rockwell's).

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Example output: AT&V

```
ACTIVE PROFILE:
E1 L3 M1 N1 Q0 T V1 X4 Y0 &A0 &C0 &E0 &H0 &P0 \A3 \N3 \V0 %C3 %E0
S00:000 S01:000 S02:043 S03:013 S04:010 S05:008 S06:002 S07:060 S08:002
...
+MS=92,1,300,56000 +FCLASS=0 +GCI=b5 +PMH=0 +PMHT=5 #CID=0 +IFC=2,2
...
STORED PROFILES: none
```

`&Wn` — Store Current Configuration

There are no stored profiles: `&W`, `&W0` and `&W1` answer `OK` and store nothing.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

`&Yn` — Profile at Power On

Accepted, changing nothing: there are no stored profiles.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 9. ATI11 Lines *(continued)*

Line	Value
`Transmit Frame Error Count`	The information frames among them sent again.
`Receive Frame Count`	The frames received intact.
`Receive Frame Error Count`	The frames received with a bad FCS.
`Retrains by Local Modem, Retrains Requested by Remote Modem`	Retrains, ours and the other modem's.
`Rate Renegotiation by Local/Remote Modem`	Rate renegotiations (`V.34`, `V.90`, `V.92`).
`Call Duration`	From `CONNECT`, *h*:*mm*:*ss*.
`Call Termination Cause`	`Local hangup` (`ATH`, `ATZ`, DTR, a character while connecting), `Remote cleardown` (the other modem's LAPM DISC or MNP LD), `Line dropped` (the SIP call ended), `Carrier lost`, `No dial tone`, `Busy`, `No answer`, `Training failed`, `Protocol failure` (no error correction where it is required, LAPM's errors), `No carrier`.

Example output: ATI11

```
Description                          Status
-----------                          ------
Last Connection                      V34
Initial Transmit Carrier Rate        33600
...
Call Duration                        0:00:10
Call Termination Cause               Local hangup
```

The digital modem has no view of the line's digital loss or robbed-bit signalling, so on `dmodemd`'s `SIP` line those two lines are not there; the analogue modem (`dmodemd -a`, `-usb`) measures them.

`+GMI`, `+GMM`, `+GMR` — Identification

These commands give the manufacturer (`D-Modem project`), the model (`D-Modem`) and the revision (`1.0.0`, the build's version). With `?` after them (`AT+GMI?`) they give the same.

Result code	When
`OK`	Always.

`+GCAP` — Capabilities

This command lists the command sets the modem has: `+GCAP: +FCLASS,+MS`.

Result code	When
`OK`	Always.

`#UD` — Unimodem Diagnostics

This command gives the last call's statistics (or this one's), as `ATI11` has them, in the form that Microsoft's Unimodem Diagnostics Command Reference Specification (version 1.0) gives, for Windows' modem log and the programs that read it: a line `DIAG <2A4D3263 key=value>` for each key there is a value for, both in hexadecimal. Before any call: `0=10`, `1=0` and `60=1`. `ATZ` clears the log; if it ended a call, the log then holds only that: `0=10`, `1=0` and `60=52` (the DTE's reset). The measures only some modulations have (`V.8`'s `CM` and `JM`, the `NMSE`, the far echo, `V.34`'s `INFO` bits) are left out for the others.

Result code	When
`OK`	Always.

Table 10. #UD Keys

Key	Value
`0`	The format (10).
`1`	The call's setup.
`2, 3`	The call's mode (data or fax) and the DTE interface's (async, or `V.80`'s transparent or framed synchronous mode).
`4, 5`	`V.8`'s `CM` and `JM`, the octets in quotes.
`10–12`	The receive and transmit levels, the noise.

Table 12. +TMO High-Level Objects *(continued)*

Tag	Objects
2E	`CallProgress`: how far the call got, why it ended (`V.59`'s codes).
2F	`ErrorControl`: the protocol, its frame sizes and windows, its frames and errors, the throughput.
30	`Compression`: what was agreed, each way, its dictionary and string sizes.
31	`DTEDCE`: the flow control, framing, rate, throughput and characters lost.

Example: AT+TMO V34 txDataHistory, AT+TMO 2E

```
txDataHistory{33600;31200}
OK
2E01{40}
2E02{90}
OK
```

4 Identification and Diagnostics

`I`*n* — Identification and Statistics

This command reports what the modem is, its settings and the last call's statistics.

Result code	When
`OK`	If *n* = 0–7 or 11.
`ERROR`	Otherwise.

Table 8. I*n* Commands

Command	Function
`I0`	The product: `D-Modem`.
`I1, I2`	The product and version (`D-Modem, 1.0.0`), then the maker, `D-Modem project`.
`I3`	The modem's name, then its line's kind (`dmodemd`: `dmodem0`, `SIP`).
`I4`	S0 to S23, eight a line (`s00=000 s01=000 ...`).
`I5`	`Stored Profile 0:` (there are none).
`I6`	The last call's statistics, as USRobotics' link diagnostics (the same as `I11`).
`I7`	The country: `Country: USA`.
`I11`	The last call's statistics, as Conexant's modems give them.

`ATI11` lists the last call's statistics: in online command state those of the call that is up, after a call (dialed or answered, connected or not) its own, until the next one; before any, `No Previous Call`. `ATI6` and `AT&V1` give the same lines, and `AT#UD` the same figures in Unimodem's form. A line whose value the call does not have is left out: the rates and what follows for a call that never came up, the frame counts without error correction, the coding lines without `V.34`, the measurements a modulation does not make, the cause while the call is up.

Table 9. ATI11 Lines

Line	Value
`Last Connection`	The modulation it came up with (`V34`, `V90`, `V92`, `V91`, `V32bis`, ...), `Fax`, or `None`.

Table 9. ATI11 Lines *(continued)*

Line	Value
`Initial Transmit/Receive Carrier Rate`	The rates (bit/s) at `CONNECT`.
`Final Transmit/Receive Carrier Rate`	The rates now, or at the end (after renegotiations).
`Symbol Rate (Tx/Rx)`	Each direction's symbols a second (`V.22` to `V.92`, the clear channel).
`Carrier Frequency (Hz, Tx/Rx)`	Each direction's carrier, `PCM` for a `V.90`, `V.91` or `V.92` `PCM` direction.
`Protocol Negotiation Result`	`LAPM`, `MNP` or `NONE`.
`Data Compression Result`	`V44`, `V42bis`, `MNP5`, `ZSTD` or `NONE`.
`Trellis Code, Nonlinear Encoding, Precoding, Shaping (Tx/Rx)`	`V.34`'s coding each way (`16S-4D`, `ON`, `OFF`; – for a `PCM` direction).
`Estimated SNR (dB)`	The receiver's signal to noise ratio (`V.34`, `V.90`'s upstream, `V.32bis`, `V.29`, `V.33`, `V.27ter`, `V.wb`).
`Receive/Transmit Signal Power Level (dBm0)`	The line's mean power each way while the call was up.
`Round Trip Delay (ms)`	As `V.34`'s phase 2 (or `V.cvm`'s start-up) measured it.
`Near/Far Echo Level (dBm0)`	The echoes of our signal the canceller removes (`V.34`, `V.90`, `V.92`).
`Carrier Frequency Offset (Hz)`	How far the far end's carrier is from ours (`V.22` to `V.92`).
`Timing Offset (ppm)`	How far the far end's clock is from ours (`V.32bis` to `V.92`).
`PCM Encoding`	The network's law, `A-law` or `u-law` (`V.90`, `V.92`).
`Digital Loss (dB), Robbed Bit Frames`	The network's pad and the frames of six whose bit it robs (`None`, 6, ...), as the analogue modem's `DIL` found them (Section 7).
`Characters Sent/Received`	The `DTE`'s.
`Transmit Frame Count`	`LAPM`'s or `MNP`'s frames sent.

Table 10. #UD Keys *(continued)*

Key	Value
13	The `NMSE` (the receiver's error over the points' least distance, in hundredths).
14–17	The echoes' loss, the far echo's delay (ms), the round trip delay.
18	`V.34`'s `INFO` bits (its Table 5).
20, 21	The modulation (`V.58`'s codes).
22–27	The symbol rates, the carriers, the first rates.
30	The carrier's losses (20 ms or more that came back).
31–33	The renegotiations, the retrains asked for and granted.
34, 35	The last rates.
40, 41	The error control and its frame size.
42, 43	The error control's timeouts and `REJ`s or `SREJ`s (`MNP`: the acknowledgements that ask for frames again).
44, 45	The compression (`V.58`'s codes; 80 `MNP 5`, 81 `V.44`, 82 zstd) and `V.42bis`' dictionary size.
50, 51	The flow control (`&H`).
52–59	The characters (54, 55: those lost for a full buffer) and frames.
60	How the call ended (`V.58`'s *callCleared*; 2 while it goes on).

Example output: AT#UD

```
DIAG <2A4D3263 0=10>
DIAG <2A4D3263 1=7>
DIAG <2A4D3263 10=0C>
...
```

`+TMO` — V.59 Diagnostics

This command gives `V.59`'s managed objects: the facts about the last call (or this one), from its start until the next call starts, as `V.250`'s `+TMO` reads them. Each object has a name and a four-digit hexadecimal tag: the first two digits its high-level object (`V34Diag` 0C, `CnxDiag` 2C, ...), the last two the object within it. An object that does not apply to the last call is `NULL`. Blanks between the parts may be left out; case does not matter.

Result code	When
`OK`	After the objects asked for.
`ERROR`	For a name or tag the modem does not have, or `+TMO` alone before any other.

Table 11. +TMO Forms

Command	Function
`+TMO`	The last `+TMO` again.
`+TMO` *`level`*`=?`, `+TMO` *`level`*`N=?`	The objects there are, as tags (`N`: as names): 0 all, 1 the high-level ones, 2 the mid-level ones, 3 the low-level ones; 4 gives `+TMO: (0,1)` (names and tags both).
`+TMO` *`tag`*	A four-digit tag: that object, as *`tag`*`{`*`value`*`}`; two digits: every object of that high-level object.
`+TMO` *`name`* `[All]`	An object by name, or every object of a high-level one (`V34Diag`, or `V34` alone), as *`name`*`{`*`value`*`}`. A modulation's object by name alone is the last call's modulation's.
`+TMO` *`high`* *`name`*	That high-level object's object (`+TMO V34 rxDataHistory`).

Table 12. +TMO High-Level Objects

Tag	Objects
00	`ModemIdentity`: the maker, model, version, `V.59`'s version (1), what the modem does (a bit string, in hexadecimal).
01	`ModemCapability`: each modulation's and protocol's mode (0 not there, 1 the preferred, 2 enabled, 3 disabled), the others there are.
02	`ModeSelected`: the call's modulation.
07	`V8Diag`: `V.8`'s `CM` and `JM` (their octets in hexadecimal), its result.
09–23	Each modulation's: its mode, the transmit and receive levels and the noise (dBm0), the symbol rates and carriers (`V.59`'s codes), the rates during the call, the echoes, the round trip delay, `V.90`'s and `V.92`'s robbed bits, digital loss and `G.711` law.
26	`NSMDiag`: a modulation `V.59` has none for (K56flex, ...).
2C	`CnxDiag`: the carriers, carrier losses, renegotiations, retrains, the call's length, whether we called.

`#CID, +VCID` — Caller ID

This command turns caller ID on or off. With it on, the modem tells the `DTE` who is calling: after the first `RING` it sends the date and time, the caller's number and, when there is one, the caller's name, a line each. `+VCID` is another name for the same command.

Caller ID works on a `SIP` call and on a telephone line. When the caller has withheld the number or name, the modem sends `P` (private) in its place; when it is not known, `O` (out of area). Value 2 sends the whole caller ID message in hexadecimal instead, for programs that decode it themselves.

Syntax:

- `AT#CID=n (or AT+VCID=n)`
- `AT#CID?`
- `AT#CID=? (gives 0,1,2)`

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

Table 18. #CID=*n* Values

Value	Function
0	Off (*default*).
1	The fields: `DATE =`, `TIME =`, `NMBR =`, `NAME =`, a line each.
2	The whole message in hexadecimal: `MESG =`.

Example output: AT#CID=1, then a call

```
RING

DATE = 1030
TIME = 1043
NMBR = +15551234
NAME = Alice Smith

RING
```

`&Tn` — Loop Tests

`V.54`'s loop tests, with Hayes' numbers. Loop 3 runs offline on a leased line (`&L`) with a 4-wire modulation (`+MS=V29`, `V33` or `V27TC`): the modem's signal goes back into its own receiver and the line is silent. Loop 2 at the far end is asked for in band (`V.54`'s `PREP`, `ACK` and `TERM`), from online command state on a call with no error correction (`\N0`). The self-test sends and checks `V.52`'s 511-bit pattern. `S18` above 0 ends a test after that many seconds.

Result code	When
`CONNECT`	For `&T1` and `&T6`: the `DTE`'s data comes back to it.
`OK`	For the others; `&T0` after `&T7` or `&T8` first gives the self-test's bit errors, three digits (255 at most).
`ERROR`	For `&T2` and *n* over 8, outside the state the test needs, or when the far end does not answer `&T6` or `&T7`.

Table 13. &T*n* Commands

Command	Function
`&T0`	Ends the test (`&T` the same).
`&T1`	Loop 3, the local analogue loop.
`&T3`	Loop 2 here: the far end's data back to it.
`&T4`	The far end may ask for loop 2.
`&T5`	The far end may not ask for loop 2 (*default*, again after `ATZ`).
`&T6`	Loop 2 at the far end, the remote digital loop.
`&T7`	Loop 2 at the far end with the self-test.
`&T8`	Loop 3 with the self-test.

`+T...` — Loop and Error Rate Tests

`V.250`'s test commands, made from `V.58`'s test objects: the loop tests of `&T`, an error rate test round them, and their status. Each also takes `?` (its value) and `=?` (its range).

5 Dialling, Answering and Hanging Up

`Dstring` — Dial

This command takes the modem off hook and dials the number in *string*, then calls. The digits and `* # A B C D` are dialed as `DTMF`, each `S11` ms long with `S11` ms between them; `T` and `P` choose tone or pulse dialing (`S16`) for the rest of the call. Pulses are dialed on an analogue line (`dmodemd`'s `-a` and `-usb`), with the country's break, make, gap between digits and digit pattern (`+GCI`; `* # A B C D` skipped); a line that cannot pulse, such as `SIP`, dials them in `DTMF`. With `X2` or `X4` the modem waits for a dial tone first. In `dmodemd` the dial string is the `SIP` call's number (Section 20).

Table 15. Dial Modifiers

Modifier	Function
`0–9 * #` `A–D`	Dialed as `DTMF`, or as pulses for the digits with `P`.
`L`	Dial the last number again (`Dialing number...` first). Valid only as the first character of the string; `ERROR` while a call is held after `;`.
`T`	Tone dialing (`DTMF`) from here on.
`P`	Pulse dialing from here on.
`,`	Pause for `S8` seconds.
`W`	Wait for a dial tone (1 s of it): `NO DIALTONE` if none comes in `S7` seconds.
`@`	Wait for the remote end's ringing (0.5 s of sound at least), then for 5 s of quiet after it: `NO ANSWER` if that does not come in `S7` seconds.
`;`	End the dial string: once the number is dialed the modem is back in command state, off hook with the call kept, and says `OK`; the rest of the line runs then.
`S=n`	A stored number: `ERROR` (there are none).
`! $`	Hook flash and bong tone: skipped (a `SIP` call has no hook to flash).

Other characters are skipped. After a `;` the call waits, silent, for the `DTE`: `ATD`*more* dials more on it (as `DTMF`; a `;` again holds it again), `ATD` alone or `ATO` starts the calling modem's handshake, `ATA` the answering modem's, and `ATH` hangs up (`OK`, then `NO CARRIER`).

Result code	When
`CONNECT`	When the call is up.
`OK`	After a `;`.
`BUSY`, `NO ANSWER`, `NO DIALTONE` or `NO CARRIER`	When it is not.
`ERROR`	For `S=n`, or `L` with no number to dial.

`P` — Select Pulse Dialing

This command sets pulse dialing for the next dial (`S16=0`). See `ATD`.

Result code	When
`OK`	Always.

`T` — Select Tone Dialing

This command sets tone (`DTMF`) dialing for the next dial (`S16=1`, the *default*). See `ATD`.

Result code	When
`OK`	Always.

`&Pn` — Pulse Dial Make/Break Ratio

The modem keeps the value in `S28`. The country's table sets the pulse times.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

Result code	When
`OK`	When the command is done: for +TRDL=1 when the far end loops, for +TAL=1 when loop 3 runs, both in online command state (not `CONNECT`: `ATO` for the DTE's data); for +TRDL=0 when the TERM has gone.
`ERROR`	Outside the state the test needs, when the far end does not loop, or for a value it does not have.

Table 14. +T Commands

Command	Function
`+TERDL=`*n*	1: the far end may ask for loop 2 (`&T4`); 0 it may not (*default*).
`+TLDL=`*n*	1: loop 2 here, from online command state (`&T3`); 0 ends it.
`+TRDL=`*n*	1: loop 2 at the far end, from online command state (`&T6`); 0 ends it.
`+TAL=`*n*`[,`*band*`]`	1: loop 3, offline on a 4-wire leased line (`&T1`); 0 ends it. *band* is taken and ignored.
`+TTER=`*type*`,`*len*`,`*blocks*`,`*pattern*	The error rate test on the loop that runs (loop 2 at the far end, loop 3), from online command state: `V.52`'s 511-bit pattern (*pattern* 1, the only one) for *blocks* blocks of *len* bits (1–65535 each); *type* 1 bit errors, 2 block errors, 3 both (both are counted). +TTER=0 stops it. +TTER? gives the type running (0 none), *len*, the blocks left and the pattern.
`+TNUM?`	The last error rate test's bit and block errors, `+TNUM: `*bits*`,`*blocks* (65535 at most).
`+TALS?`	Loop 3: 0 none, 3 running.
`+TDLS?`	Loop 2 here: 1 none, 3 ours (+TLDL, `&T3`), 4 the far end's.
`+TRDLS?`	Loop 2 at the far end: 1 none, 4 running.
`+TSELF=`*n*	The self-test (0 full, offline only; 1 partial): the loop tests' pattern generator, checker and recognisers check themselves.
`+TRES?`	Its result: 0 none since `ATZ`, 1 pass, 2 fail.

`+TE140, +TE141,`
`+TEPDL, +TEPAL,`
`+TMODE, +TADR`
	0 alone: no circuits 140 and 141, no front panel, point to point, no `V.54` address.

`%E`*n* — Line Quality Monitor

The modem keeps the value in S39. Its modulations retrain and renegotiate their rate by themselves.

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

`A` — Answer

This command takes the modem off hook to answer a call. After S92 seconds it sends the answering modem's tones. In `dmodemd` the call that rings (`RING`) is answered. On a call held after a dial string's `;`, `ATA` answers that call: `ATD`*number*`;A` dials, then answers (`V.250`'s call-back).

Result code	When
`CONNECT`	When the call is up.
`NO CARRIER`	When no call comes up.
`ERROR`	If there is no call to answer.

`H`*n* — Hook Control

This command puts the modem on or off hook. `H` or `H0` hangs up a call. In voice mode `ATH` leaves the line and goes back to data mode (Section 18).

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	For another number, or when the hook cannot change.

Table 16. H*n* Commands

Command	Function
`H0`	On hook (hang up).
`H1`	Off hook.

`O`*n* — Return to On-Line Data Mode

This command takes the modem back online after the escape (Section 19). On a call held after a dial string's `;` it starts the calling modem's handshake (as `ATD`). `O1` and `O3` work in `V.34`, `V.90` and `V.92` calls; in the others they only go back. A retrain or renegotiation counts as the local modem's in `ATI11`.

Result code	When
`CONNECT`	When the modem is back online.
`ERROR`	When there is no call to go back to, or for *n* = 2 or over 3.

Table 17. O*n* Commands

Command	Function
`O0`	Back online (*default*).
`O1`	Back online, then retrain (`V.250`'s).
`O3`	Back online, then renegotiate the rate (Rockwell's).

`Y`*n* — Long Space Disconnect

Other modems use `Y` for long space disconnect. The modem keeps the value in S161 and does nothing more.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

`%B, *B` — Blacklisted Numbers

These commands list the numbers barred from dialing. None are kept.

Result code	When
`OK`	Always.

+GCI — Country of Installation

This command sets the country by its `T.35` code in hexadecimal (`+GCI:b5` is the USA). The country sets its dial pause (S8) and hook flash (S24) times, its pulse dialing and its tones. `dmodemd`'s `-c` option names the country it starts with (`-countrylist` lists them): one of the table's, or any of the 192 of the `ITU`'s list of the tones of national networks (after `E.180`), by its name there in capitals or the table's (`-c FIJI`, `-c UNITED_KINGDOM`: `UK`). The table's countries have their own dialling settings; the others the USA's (`ATI7`: `Country:  FIJI (USA line)`), and all of them their own busy and congestion tones. `ATZ` keeps the country.

Syntax:

- `+GCI=`*h*
- `+GCI?  (gives +GCI:b5)`
- `+GCI=?  (lists the codes)`

Result code	When
`OK`	Always.
`ERROR`	For a code that is not in the table.

7 Modulation

+MS — Modulation Selection

This command sets the modulation the modem calls or answers with, whether it may fall back (*automode* 1, as `N1`; `S31`), and the rates it may connect at, *min* to *max* bit/s. *mod* is a number or `V.250`'s name for it, in either case (`AT+MS=V34` is `AT+MS=34`). A parameter left out keeps its value: `AT+MS=,0` only turns automode off. In `V.34`, *max* caps both directions' rates (sent in its `MP` sequences).

Syntax:

- `+MS=[`*mod*`][,[`*automode*`][,[`*min*`][,[`*max*`]]]]`
- `+MS? (gives 92,1,300,56000; 200,1,300,56000 at 48000 Hz and more)`
- `+MS=? (gives (21,22,23,26,27,29,122,32,132,232,61,33,34,103,18, 45,110,5,12,212,201,202,208,56,90,91,92,150,200,64),(0,1), (300-56000),(300-56000), the modulations the program has)`

Result code	When
`OK`	Always.
`ERROR`	For a modulation the modem has not, *automode* over 1, *min* over *max*, or a rate outside 300 to 56000.

Table 22. +MS Modulations

mod	Name	Standard	Description
21	`V21`	`ITU V.21`	300 bit/s `FSK`.
23	`V23C`	`ITU V.23`	1200/75 bit/s `FSK`.
103	`B103, BELL103`	Bell 103	300 bit/s `FSK`.
22	`V22`	`ITU V.22`	1200 bit/s `DPSK`.
212	`B212, BELL212A`	Bell 212A	1200 bit/s `DPSK`.
122	`V22B`	`ITU V.22bis`	2400 bit/s, `V.22` as its fallback.
32	`V32`	`ITU V.32`	4800 and 9600 bit/s.
132	`V32B`	`ITU V.32bis`	4800 to 14400 bit/s.
232	`V32T, V32TERBO`	`V.32terbo`	`V.32bis` and 16800, 19200 bit/s (AT&T's; D-Modem's number).

Technical details:

- `DATE` is the month and day (MMDD), `TIME` the hour and minute (HHMM).
- In voice mode (`+FCLASS=8`) the lines come as `V.253`'s complex event report, between `<DLE>X` and `<DLE>.`; in the other modes after an empty line.
- On a telephone line the modem listens for caller ID after the first ring (North America's way, and `ETSI`'s after a short ring). On a line it hears on hook (`ALSA`, a pipe, the pico-serial Line) it also listens before the ring (`ETSI`'s after its alerting tone, the UK's, `DTMF` caller ID).
- It accepts Bell 202 `FSK` and `V.23 FSK` (`ETSI` EN 300 659-1, BT SIN 227) alike, and `DTMF` caller ID: `A` and the number, or `B` (or `D`) and a reason it is absent (`10` private, as `P`; others `O`), up to `C`, `#` or a pause.
- In voice mode with the line taken it accepts Type II caller ID too, a waiting call's caller: after the `CPE` alerting signal (2130 and 2750 Hz) it answers with `DTMF A` (GR-30's, a `CPE` without `ADSI`) and reports the `FSK` message that follows as a packet, the line's audio to the `DTE` silent meanwhile.
- A `SIP` call (`dmodemd -L`) has its caller ID from the INVITE: the caller's P-Asserted-Identity, else its From (the `URI`'s user part as `NMBR`, the display name as `NAME`), the date and time the host's. A withheld number or name (the Privacy header's `id`, `user` or `header`, or an anonymous caller) is `P`; with value 2 the modem sends the `MDMF` message it made of them.

V.cvm. `VCVM` is the proposed Recommendation of `docs/v_cvm.pdf`, for calls whose speech path is a mobile codec's (`AMR`, AMR-WB), which carry neither `V.34` nor `V.32`: `V.8` with `V.22` and `V.21` as the fallback, then a probing of the codec and a mode loaded for it. The start-up takes 15 to 21 s; `S7` should allow for it. The connection runs `LAPM`. `V_CVM`, its old name, is still accepted.

V.wb. `VWB` is D-Modem's own mode for a dry copper pair between two of its modems (pico-serial's WB stick, a codec at 96 kHz): discrete multitone, the caller on the lower tones and the answerer on the upper, each tone loaded from its measured `SNR`; see `docs/dsp/vwb.md`. It runs at 48000 Hz and more only, where it is the only mode and the *default* (`ATZ` sets it again there), as a leased line (`&L1`); the rates of `+MS` do not limit it.

Clear Channel. `CLEAR` is a clear 64 kbit/s digital channel, such as an `ISDN` B-channel's data call (a `SIP` call in CLEARMODE, `RFC` 4040, or a `PBX`'s call with an unrestricted digital bearer): no modulation, each of the channel's octets 8 bits of the link's, `CONNECT 64000` both ways. It starts at once on `ATD` and `ATA`, with no call progress and no answer tone. Each octet's first bit is its most significant (`I.460`, `RFC` 4040). A line that says its call is clear (`dmodemd`'s `SIP` for a CLEARMODE call, `app_dmodem` for an `ISDN` data call) runs it whatever `+MS` says. What runs on the channel is `+B2`'s, `X.75` by *default*, or `V.110`'s rate adaption with `+CBST`. See `docs/isdn.md`.

6 Speaker, Line and Country

L*n* — Speaker Volume

This command sets the speaker's volume (S30). A SIP line has no speaker: the setting is kept. A SmartUSB56 with `dmodemd -usb-buzzer` sounds its buzzer while the speaker is on, at any volume but `L0`.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

Table 19. L*n* Commands

Command	Function
`L0`	Off.
`L1`	Low volume.
`L2`	Medium volume.
`L3`	High volume (*default*).

M*n* — Speaker Control

This command sets when the speaker is on (S29).

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

Table 20. M*n* Commands

Command	Function
`M0`	Speaker always off.
`M1`	Speaker on until the call is up (*default*).
`M2`	Speaker always on.
`M3`	As `M1` (on other modems: not while dialing).

&J*n* — Auxiliary Relay

Accepted, changing nothing: there is no relay.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

&L*n* — Leased Line

This command sets whether the line is a dialled one or a leased pair or circuit, with no exchange: no dial tone, ringing or hook (S47). On a leased line `ATD` (any number, or none) calls at once, listening only for the answer tone; `ATA` answers; with S0 above 0 the modem answers by itself, with no ring, whenever it is idle. A call the line ends (`NO CARRIER`) starts again by itself 2 s later, dialled or answered as before; `ATH`, DTR, `ATZ` or `&L0` stop that until the next command line.

Every voice band modulation of `+MS` runs on `&L1`. On `&L2` the 4-wire modulations (`+MS=V29`, `V33`, `V27TC`) run as duplex pumps on a continuous carrier, each end starting at once with its own synchronizing signal; the others run as on `&L1`. With `S48=1`, on a call with no error correction, the modem does `V.13`'s simulated carrier control: between the DTE's bursts of data the line carries `V.13`'s idle pattern, which the far end treats as carrier detect OFF, and each burst turns it ON again. A line at 48000 Hz and more (`dmodemd`'s `-r 96000`) is always leased: its mode is `+MS=VWB`, and its start-up and retrains take S7 seconds at most.

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

Table 21. &L*n* Commands

Command	Function
`&L0`	A dialled line (*default*).
`&L1`	A 2-wire leased line.
`&L2`	A 4-wire leased line.

Table 22. +MS Modulations (*continued*)

mod	Name	Standard	Description
61	`V61`	`ITU V.61`	4800 bit/s with speech, 4800 to 14400 bit/s without (`+ASVD`; D-Modem's number).
29	`V29`	`ITU V.29`	4800 to 9600 bit/s, 4-wire leased.
33	`V33`	`ITU V.33`	12000 and 14400 bit/s, 4-wire leased.
27	`V27TC`	`ITU V.27ter`	2400 and 4800 bit/s, 4-wire leased.
26	`V26`	`ITU V.26`	2400 bit/s DPSK (alternative A), 4-wire leased.
201	`B201, BELL201`	Bell 201	2400 bit/s DPSK (`V.26`'s alternative B), 4-wire leased.
202	`B202, BELL202`	Bell 202	1200 bit/s FSK both ways, 4-wire leased.
208	`B208, BELL208`	Bell 208	4800 bit/s 8-phase DPSK, 4-wire leased.
34	`V34`	`ITU V.34`	2400 to 33600 bit/s, after `V.8`.
90	`V90`	`ITU V.90`	As the digital modem, after `V.8`.
91	`V91`	`ITU V.91`	Two digital modems, after `V.8`.
92	`V92`	`ITU V.92`	`V.90` with quick connect, PCM upstream and modem-on-hold (*default*).
56	`K56`	K56flex	Not implemented: runs `V.90`.
150	`VCVM`	`V.cvm`	Data over mobile speech paths, 175 to 3500 bit/s, after `V.8` (`+CVM`).
200	`VWB`	`V.wb`	Wideband DMT on a dry pair, at 48000 Hz and more, up to 148656 bit/s each way.
64	`CLEAR`	Clear channel	64000 bit/s on a clear digital channel (an ISDN data call), no modulation (`+B2`).
18	`V18`	`ITU V.18`	Text telephone automoding (Section 15).

Table 22. +MS Modulations (*continued*)

mod	Name	Standard	Description
45	`BAUDOT`	`V.18`, TIA-825-A	5-bit text telephone: TTY, telex, RTTY (`+MBAUDOT`).
110	`EDT`	`V.18`	EDT text telephone, 110 bit/s.
5	`CONTACTID`	SIA DC-05	Contact ID alarm reports: the answerer is the receiver, the caller the panel (Section 16).
12	`SMS`	ETSI ES 201 912	Fixed-line text messages: the answerer receives, the caller sends (Section 17).

The 4-Wire Modulations. `V.29`, `V.33`, `V.27ter`, `V.26` and Bell 201, 202 and 208 are the 4-wire leased line modulations (`&L2`): each end sends on its own pair, on a carrier that stays on. `V.26` and Bell 201 send their data unscrambled, as those modems do, with the phase changes of `V.26`'s alternative A or B; Bell 208 sends `V.27ter`'s 4800 bit/s signal, the modulation `V.27` took from it; Bell 202 is 1200 Hz mark and 2200 Hz space at 1200 bit/s each way. The DPSK ones start with `V.27ter`'s training as their synchronizing signal, which another D-Modem needs: an old modem of these kinds that sends none is not found. *max* sets the rate each end sends at, *min* the lowest it comes down to: a receiver finds the far end's rate by itself, an end faster than the far end comes down to its rate, and one that loses its equalization twice in 20 s at its own rate drops to the next rate down. A receiver that loses its equalization sends its synchronizing signal again and gets the far end's back (`V.29` 10, `V.33` 9).

V.90, `V.91` and `V.92`. `V.90` and `V.92` are there as the *digital* modem (the server's side, as on a `G.711` VoIP line), on an 8000 Hz `G.711` line: analogue `V.90` and `V.92` modems calling connect at up to 56000 bit/s downstream. 90 and 92 offer `V.91` too, which two digital modems use instead, and `V.34` for the others; on a line at another sample rate they are `V.34`. With automode `V.8` and the `V.34` modem hand over to `V.32bis` (and its `V.22bis` and `V.22`) for a modem without `V.34`, and the answering modem tries `V.23` and Bell 103 for one without `V.8`.

On an Analogue Line. On an analogue telephone line (`dmodemd`'s `-a` and `-usb`) the modem is the call's analogue end instead: `V.90` and `V.92` run as the analogue modem against a `V.90` or `V.92` server (at 9600 Hz; see `docs/hardware.md`), and `V.91` and `CLEAR`, which need a digital line, run as `V.34`.

B*n* — Communication Standard Setting

`ATB` with no number sets `V.22` at 1200 bit/s only, without automode: the same as `AT+MS=22,0,1200,1200`. Other modems use `B0` to `B3`, `B15` and `B16` to choose between the `ITU` and the Bell modulation at 1200 and 300 bit/s. The modem accepts them and changes nothing: the call's negotiation and `+MS` chose.

Result code	When
`OK`	If *n* is left out, or *n* = 0–3, 15 or 16.
`ERROR`	Otherwise.

Table 23. B*n* Commands

Command	Function
`B`	`V.22` at 1200 bit/s, no automode.
`B0, B1`	Accepted (other modems: `V.22` or Bell 212A at 1200 bit/s).
`B2, B3`	Accepted (other modems: `V.23`'s reverse channel).
`B15, B16`	Accepted (other modems: `V.21` or Bell 103 at 300 bit/s).

N*n* — Automode

This command sets whether the modems may find their best common modulation from the one `+MS` chose down (`S31`). It is the same as `+MS`'s second parameter.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 24. N*n* Commands

Command	Function
`N0`	Only the modulation `+MS` chose.
`N1`	Automode: that modulation or the best common one below it (*default*).

C*n* — Carrier Control

The modem accepts `C1` so that software that sends it works. It does not have `C0`, which on some other modems sends no carrier (receive only).

Result code	When
`OK`	If *n* = 1.
`ERROR`	Otherwise (`C` alone is `C0`).

Table 25. C*n* Commands

Command	Function
`C0`	Carrier always off (not supported).
`C1`	Normal carrier switching (*default*).

&B*n* — V.32 Auto Retrain

Other modems use `&B` for `V.32`'s retrain (USRobotics': the `DTE`'s rate fixed). The modem accepts it and changes nothing: its modulations retrain by themselves.

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

&G*n* — V.22bis Guard Tone

Accepted, changing nothing: the modem sends no guard tone.

Result code	When
`OK`	If *n* = 0–2.
`ERROR`	Otherwise.

7.2 V.92 Call Waiting and Modem-on-Hold

`+PCW`, `+PIG` — V.92 Call Waiting and `PCM` Upstream

`+PCW` (0 to 2, *default* 2) is `V.92`'s call waiting: none is detected here. `+PIG` (0 or 1, *default* 1) is `V.92`'s `PCM` upstream, an analogue modem's setting. Kept.

`+PMH` — Modem-on-Hold Enable

This command sets whether the other modem may put the call on hold.

Syntax:

- `+PMH=`*n*
- `+PMH?` (gives `+PMH: 0`)
- `+PMH=?` (gives `+PMH: (0-1)`)

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 27. +PMH Values

Value	Meaning
0	The other modem may put the call on hold (*default*).
1	It may not.

`+PMHT` — Modem-on-Hold Timer

How long a hold the modem grants: the code of `V.92`'s Table 33, 0 to 13; 0 grants none.

Syntax:

- `+PMHT=`*n*
- `+PMHT?` (gives `+PMHT: 5`)
- `+PMHT=?` (gives `+PMHT: (0-13)`)

Result code	When
`OK`	If *n* = 0–13.
`ERROR`	Otherwise.

`+A8E` — V.8 and V.8 bis Controls

`V.251`'s controls of `V.8` and `V.8 bis`. *v8o* and *v8a* are 1: the modem runs `V.8` itself on calls and answers (`+MS` decides whether it does). *v8cf* is 0: no call function of the `DTE`'s. *v8b* 1 starts a data call with `V.8 bis`: answering, the modem sends MRe before its answer tone and, with a caller that answers it, agrees on the data mode with it (CL, MS, ACK), then goes on with `V.8`; calling, it answers the other modem's MRe the same way. A caller without `V.8 bis` is answered with `V.8` after 3 s. Only for the modulations that start with `V.8` (`V.34`, `V.90`, `V.91`, `V.92`, `V.cvm`), not on a leased line or a clear channel. See `docs/dsp/v8bis.md`.

Syntax:

- `+A8E=[`*v8o*`][,[`*v8a*`][,[`*v8cf*`][,[`*v8b*`]]]]`
- `+A8E?` (gives `+A8E: 1,1,0,0`)
- `+A8E=?` (gives `+A8E: (1),(1),(0),(0,1)`)

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range: *v8b* 2 (the `DTE`'s own `V.8 bis`, `+A8M`), *cfrange* and *protrange*.

`V.251`'s *default* *v8b* is 1; D-Modem's is 0, since answering with `V.8 bis` keeps a caller without it waiting about 3.5 s.

`+DSVD` — Voice and Data Together

D-Modem's own: *on* 1 offers `V.70`'s digital simultaneous voice and data (`DSVD`) in `V.8 bis` (`+A8E`'s *v8b* 1), and chooses it when the other modem offers it too. The call then goes on with `V.8` and its modulation, and the link is `V.76`'s multiplexer: the `DTE`'s data goes on its data channel, opened with `V.75`'s messages, with error correction and no compression. The protocol is then `V.76` (`ATI11`, the `Protocol:` report). Where the modem has a voice path (a SmartUSB56's handset, or dmodemd's `-dsvd-audio` command), a voice channel opens beside the data, the speech coded with `G.729` Annex A, and silence with Annex B (comfort noise, the line left to the data) when both ends send `V.75`'s audio header. Without `DSVD` at the other end the call is an ordinary data call. See `docs/v70.md`.

Syntax:

- `+DSVD=[`*on*`]`
- `+DSVD? (gives +DSVD: 0)`
- `+DSVD=?  (gives +DSVD: (0,1))`

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

`+ASVD` — Analogue Voice and Data Together

D-Modem's own: *on* 1 offers `V.61`'s analogue simultaneous voice and data (`ASVD`) in `V.8 bis` (`+A8E`'s *v8b* 1), and chooses it when the other modem offers it too. The call then goes on with `V.61`'s own start-up instead of `V.8`: one carrier of `3000` (or `2800`) symbols a second carries `4800` bit/s of data with the speech around it, and data alone at up to `14400` bit/s while neither end speaks. The `DTE`'s data has `V.42`'s error correction as on any call. The speech comes from and goes to the same voice path as `+DSVD`'s (a SmartUSB56's handset, or dmodemd's `-dsvd-audio` command); without one every frame is data alone. When both ends offer `+DSVD` as well, `DSVD` is chosen. Without `ASVD` at the other end the call is an ordinary data call. `AT+MS=V61` runs `V.61` without `V.8 bis`. See `docs/dsp/v61.md`.

Syntax:

- `+ASVD=[`*on*`]`
- `+ASVD? (gives +ASVD: 0)`
- `+ASVD=?  (gives +ASVD: (0,1))`

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

`&N`*n* — Line Rate

USRobotics' choice of the line's rate: accepted, changing nothing. `+MS` sets the rates.

Result code	When
`OK`	If *n* = 0–14.
`ERROR`	Otherwise.

7.1 V.cvm

`+CVM` — V.cvm Settings

For `+MS=VCVM`: *modes*, the modes allowed, a sum of the values below; *rate*, the signal interface's rate, 8000 or 16000; *report*, 0 or 1, asks for `+CVMR` on each renegotiation. D-Modem keeps *rate* and *report* but works at 8000 Hz only, without the wideband modes, and sends no `+CVMR`. During and after a `V.cvm` call, `+CVM?` gives a second line with that call's figures: `+CVM: `*`raw tx`*`,`*`raw rx`*`,`*`coded tx`*`,`*`coded rx`*`,`*`symbol ms`*`,`*`tones tx`*`,`*`tones rx`*`,`*`code rate`*`,`*`renegotiations`*`,`*`erasures`* (`+CVM: 1600,1600,1400,1400,20.0,16,16,7/8,0,0`; the tones are 0 in the parametric mode). The call's `CONNECT` has its coded receive rate and `/CVM`: `CONNECT 1200/CVM`, with `\V1` `CONNECT 1200/CVM/LAPM`. `S44` and `S45` hold the tones loaded and the renegotiations.

Syntax:

- `+CVM=[`*modes*`][,[`*rate*`][,[`*report*`]]]`
- `+CVM? (gives +CVM: 63,8000,0)`
- `+CVM=?  (gives +CVM: (0-63),(8000,16000),(0,1))`

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

Table 26. +CVM Modes

Value	Mode
1	The robust mode.
2	Narrowband.
4	Wideband (not supported).
8	8-DPSK.
16	The amplitude rings.
32	The parametric mode.
63	All of them (*default*).

`+PMHR` — Modem-on-Hold Request

In online command mode on a `V.92` connection, this command asks the other modem to go on hold for *s* seconds (1 to 960; 10 if left out).

Syntax:

- `+PMHR[=s]`

Result code	When
`OK`	Always.
`ERROR`	Outside online command mode of a `V.92` call, or for *s* outside 1 to 960.

`\Nn` — Error Control Mode

This command sets the error correction (S103), as Rockwell's modems have it. Each value sets `+ES` to `V.250`'s equal. `ATS103=n` is `\Nn`.

`V.42`'s detection phase decides between `LAPM` and none. The originator sends `MNP`'s `LR` when the answerer sends no `V.42` answer, and the answerer treats an `LR` (from a modem with `MNP` only, or one set so) as the start of `MNP`. With `MNP` only the originator starts with its `LR`. `MNP 5` compression goes with `MNP` (`%C`); `MNP`'s block size is `\A`'s.

Result code	When
`OK`	If *n* = 0–5.
`ERROR`	Otherwise.

Table 30. \N*n* Commands

Command	Function	+ES
`\N0`	Normal: no error correction.	`1,0,1`
`\N1`	Direct: no error correction.	`0,0,0`
`\N2`	Reliable: `V.42` (`LAPM`) or `MNP`, else hang up.	`3,2,4`
`\N3`	Auto-reliable: `V.42`, `MNP`, or none (*default*).	`3,0,2`
`\N4`	`V.42` (`LAPM`) only, else hang up.	`3,3,5`
`\N5`	`MNP` only, else hang up.	`4,4,6`

`&Mn` — Asynchronous Communications Mode

Accepted, changing nothing (other modems: asynchronous mode; USRobotics': error correction, here `\N`).

Result code	When
`OK`	If *n* = 0, 4 or 5.
`ERROR`	Otherwise.

`\An` — `MNP` Block Size

This command sets `MNP`'s block size, the longest information field it offers (S105). As other modems do, an `LR` offers 64 and the 256-octet data phase optimization.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

Table 31. \A*n* Commands

Command	Function
`\A0`	64 octets.
`\A1`	128 octets.
`\A2`	192 octets.
`\A3`	256 octets (*default*).

`-SEC` — `MNP` 10EC

Rockwell's `MNP 10EC`, and its transmit level (`1,12`): kept, *default* 0. `-SEC?` gives the values alone, without the name; `=?` gives `(0,1),(0-30)`.

`-C, -K, -Q, )M, @M, *H` — Rockwell's `MNP` 10

Rockwell's `MNP 10` and 10EC (cellular): the calling tone, `MNP` Extended Services, the `V.22bis` fallback, power levels and the link's speed. The modem has no `MNP 10`: it accepts them, changing nothing. A modem with `MNP 10EC` connects with `V.42`.

Result code	When
`OK`	For `-C0`, `-C1`, `-K0` to `-K2`, `-Q0`, `-Q1`, `)M0` to `)M2`, `@M0` to `@M30`, `*H0` to `*H2`.
`ERROR`	Otherwise.

8.1 V.80 Synchronous Access

With `+ES=6` (calling) or `+ES=,,8` (answering) the call has no error correction or compression: after `CONNECT` the modem moves the `DTE`'s octets as a synchronous bit stream, with `V.80`'s in-band commands after EM (19h) both ways (Table 9/V.80).

Table 33. +ESA Parameters *(continued)*

Parameter	Value	Meaning
	`1`	FCS-16.
	`2`	FCS-32 (`V.42`'s).
`nrzi_en`	`0`	Only 0.
`syn1, syn2`	`0-255`	The `SYN` octets (*default* 255).

`+ITF` — Transmit Buffer Thresholds

The Synchronous Access Mode's transmit buffer: the `DTE`'s octets in it, EM codes included, stop being accepted above *off*; *on* is kept. With *report_period* not 0 the modem gives `<EM><bnum>` with their number every *report_period* times 10 ms.

Syntax:

- `+ITF=[off][,[on][,[report_period]]]`
- `+ITF? (gives +ITF: 383,128,0)`
- `+ITF=?  (gives +ITF: (1-4095),(0-4094),(0-500))`

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range, or *off* not above *on*.

8 Error Control

+ES — Error Control Selection

V.250's error control selection: what the modem tries first as the originator (*orig*), what may follow when that fails (*orig_fb*), and what it accepts as the answerer (*ans_fb*). \N sets it, and it sets S103 to the \N with its values (as it was if none has them). V.80's Synchronous Mode and Frame Tunnelling Mode (5 and 7, 7 and 9) are not there.

Syntax:

- `+ES=[orig][,[orig_fb][,[ans_fb]]]`
- `+ES?` (gives `+ES: 3,0,2`)
- `+ES=?` (gives `+ES: (0-4,6),(0-4),(0-6,8)`)

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

Table 28. +ES Values

Parameter	Value	Meaning
orig	0, 1	No error correction.
	2	V.42 (LAPM) without the detection phase.
	3	V.42 with the detection phase (*default*).
	4	MNP first (no detection phase, no LAPM).
	6	V.80's Synchronous Access Mode (Section 8.1).
orig_fb	0, 1	None (*default* 0).
	2	V.42 or MNP, or hang up.
	3	V.42 only.
	4	MNP only (and first).
ans_fb	0, 1	No error correction.
	2, 3	V.42, MNP or none (*default* 2).
	4	V.42 or MNP, or hang up.
	5	V.42 only.
	6	MNP only.
	8	V.80's Synchronous Access Mode.

&Q*n* — Error Control Mode

Other modems' choice of error correction, as \N. &Q alone is &Q0. The synchronous modes are not there.

Result code	When
`OK`	If *n* = 0, 5, 6, 8 or 9.
`ERROR`	Otherwise.

Table 29. &Q*n* Commands

Command	Function
&Q0, &Q6	No error correction (\N0).
&Q5, &Q8, &Q9	V.42, then MNP, then none (\N3).

It starts in the *transparent* sub-mode: the DTE's octets go out as they are, least significant bit first, and the line's bits come back as octets; when the DTE has nothing the modem sends *syn1* and tells it <EM><under>. With +ESA's *trans_idle* 1 or 2 the receiver first hunts for the SYN, and gives the DTE nothing before it. <EM><flag> from the DTE starts the *framed* sub-mode: HDLC frames, each ended by the DTE's <EM><flag>, the modem adding the zero insertion, the flags and the FCS. It gives back each received frame's octets (without the FCS it checked) and <EM><flag> for a good one or <EM><err> for a bad one, an <EM><err> when its hunt finds a flag, and <EM><mark> for an abort or marks. <EM><mark> from the DTE goes back to transparent. EM and 99h in the data are shielded both ways (<EM><t1>, <EM><t2>; DC1 and DC3 too with DC1/DC3 flow control, +IFC); from the DTE all the codes T1 to T20 are accepted. V.34's half duplex is not there.

Table 32. V.80 In-Band Commands

Command	From	Meaning
`<EM><ecs>`	DTE	Online command state (instead of +++, which is data); the modem answers <EM><ecs>, then OK; ATO goes back.
`<EM><eot>`	DTE	Hang up: <EM><eot>, then NO CARRIER.
`<EM><rate><tx><rx>`	DTE	The highest rates to send and to receive at (+MS's *max*), for the next retrain or renegotiation (AT+MS sets both again).
`<EM><hunt>`	DTE	Restart the receiver's hunt (framed, or transparent with *trans_idle* 1 or 2); what waits for the DTE is dropped.
`<EM><resume>`	DTE	Go on after an underrun (+ESA's *framed_un_ov*).
`<EM><rtn>`, `<EM><rrn>`	DTE	Retrain, or renegotiate the rate, as ATO1 and ATO3.
`<EM><eot>`	Modem	The remote's carrier is gone (before NO CARRIER).
`<EM><rtn>`, `<EM><rrn>`	Modem	A retrain or renegotiation but the DTE's own (the remote's, or the modem's after a loss of signal).
`<EM><rate><tx><rx>`	Modem	The rates, after each of those, or when they change in the call.
`<EM><rover>`	Modem	Received octets were lost (the DTE read nothing for 2048 octets).

Table 32. V.80 In-Band Commands *(continued)*

Command	From	Meaning
`<EM><tover>`	Modem	The DTE's octets were lost (the buffer full).
`<EM><under>`	Modem	The DTE ran out of data.
`<EM><bnum>`	Modem	The transmit buffer's count, with +ITF.

+ESA — Synchronous Access Mode Settings

The Synchronous Access Mode's eight settings, each left out as it is.

Syntax:

- `+ESA=[trans_idle][,framed_idle...][,syn2]`
- `+ESA?` (gives `+ESA: 0,0,0,0,0,0,255,255`)
- `+ESA=?` (gives `+ESA: (0-2),(0,1),(0,1),(0),(0-2),(0),(0-255),(0-255)`)

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

Table 33. +ESA Parameters

Parameter	Value	Meaning
`trans_idle`	0	*syn1* sent when the DTE has nothing; the receiver's octets as they come (*default*).
	1	The same, but the receiver hunts for *syn1*.
	2	*syn1* and *syn2* sent in turn; the receiver hunts for the two (*syn1* first). Each hunt gives the DTE the octets from the SYN on, aligned on it.
`framed_idle`	0	Flags between frames (*default*).
	1	Marks between frames.
`framed_un_ov`	0	The DTE runs out in a frame: an abort (*default*).
	1	A flag (no FCS). Either way the DTE's octets are then dropped until its <EM><resume>, which <EM><unum> and their count answer (both: <EM><under>).
`hd_auto`	0	Only 0.
`crc_type`	0	No FCS (*default*).

9 Data Compression

&K*n* — Data Compression

&K0 to &K3 set data compression as on USRobotics' modems, the same as %C (S104). Rockwell's and Conexant's modems use &K for flow control: their &K4 to &K6 are accepted with no effect, and their &K3 (RTS/CTS) leaves `V.42`'s compression on.

Result code	When
`OK`	If *n* = 0–6.
`ERROR`	Otherwise.

Table 34. &K*n* Commands

Command	Function
`&K0`	No compression (%C0).
`&K1, &K2`	`V.44`, `V.42bis` or `MNP 5` (%C3).
`&K3`	`V.44` or `V.42bis`, no `MNP 5` (%C2).
`&K4–&K6`	Accepted (other modems' flow control).

+DS44 — V.44 Data Compression

`V.44` (`V.250` 6.6.2), set the same way. AT+DS=0 with AT+DS44=3 offers `V.44` alone, AT+DS44=0 `V.42bis` alone (and zstd, unless +DSZ=0), both 0 no compression, as AT%C0. zstd has the directions either has.

Syntax:

- `+DS44=[`*`direction`*`][,[`*`neg`*`][,[`*`capability`*`][,...]]]`
- `+DS44?  (gives +DS44: 3,0,0,4096,4096,255,255,12288,12288)`
- `+DS44=?  (gives the ranges below)`

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

Table 37. +DS44 Parameters

Parameter	Values	Meaning
`direction`	`0–3`	As +DS's (*default* 3).
`neg`	`0, 1`	1: hang up when the link comes up without `V.44`, or zstd in its place, in those directions.
`capability`	`0, 1`	Kept: the stream method is used whatever it says.
`codewords tx, rx`	`256–65535`	The largest codewords, each way; at most 4096.
`strings tx, rx`	`32–255`	The longest strings; at most 255.
`history tx, rx`	`512–65535`	The history, each way; at most 12288.

Of each pair the smaller is used both ways.

+DSZ — zstd Data Compression

D-Modem's own: *on* 1 offers zstd (Zstandard, `docs/v_cvm.pdf`'s Annex F) in `V.42`'s `XID`, beside `V.42bis` and `V.44`, whenever compression is on (%C); *dict* 1 offers it with `V.cvm`'s dictionary. A call has zstd only when the other modem offers it too (another D-Modem): `CONNECT` then ends in `/ZSTD`, and the compression result is `ZSTD` (70 with V0); other modems go on with `V.44` or `V.42bis`. 0 offers `V.44` and `V.42bis` alone.

Syntax:

- `+DSZ=[`*`on`*`][,[`*`dict`*`]]`
- `+DSZ? (gives +DSZ: 1,1)`
- `+DSZ=?  (gives +DSZ: (0,1),(0,1))`

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

&D*n* , &F*n* , &R*n* , &S*n* — `DTR`, Factory Defaults, `RTS`, `DSR`

The modem accepts these, with or without a digit, and does nothing: ATZ gives the *defaults*, and `dmodemd` hangs up when the `DTE` drops `DTR` whatever &D says (Section 19).

Result code	When
`OK`	Always.

F*n* — On-Line Data Character Echo

The modem accepts F1 so that software that sends it works. Other modems' F0 echoes the data online; this modem never does.

Result code	When
`OK`	If *n* = 1.
`ERROR`	Otherwise (F alone is F0).

\J*n* — Rate Adjust

Other modems' `DTE` rate following the line's: accepted, changing nothing.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

\K*n* — Break Control

What a break does on other modems: accepted, changing nothing.

Result code	When
`OK`	If *n* = 0–5.
`ERROR`	Otherwise.

%C*n* — Data Compression Control

This command sets data compression (S104). It needs error correction (\N). V.42 offers V.44 and V.42bis both, and the other modem chooses; +DS, +DS44 and +DSZ set them in detail. &K sets the same.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

Table 35. %C*n* Commands

Command	Function
%C0	No compression.
%C1	MNP 5 (with MNP).
%C2	V.44 or V.42bis (with V.42's LAPM).
%C3	Either, as the call's error correction has it (*default*).

+DS — V.42bis Data Compression

V.42bis (V.250 6.6.1), within %C's compression.

Syntax:

- +DS=[*direction*][,[*neg*][,[*dict*][,[*string*]]]]
- +DS? (gives `+DS: 3,0,2048,34`)
- +DS=? (gives `+DS: (0-3),(0,1),(512-65535),(6-250)`)

Result code	When
`OK`	Always.
`ERROR`	For a value outside its range.

Table 36. +DS Parameters

Parameter	Values	Meaning
`direction`	0-3	The DTE's directions: 0 none, 1 what it sends, 2 what it receives, 3 both (*default*).
`neg`	0, 1	1: hang up (`NO CARRIER`) when the link comes up without compression in those directions. *Default* 0.
`dict`	512-65535	The largest dictionary V.42's XID offers; at most 2048, the *default*.
`string`	6-250	The longest string offered; at most 34, the *default*.

---

10 DTE Interface and Flow Control

+IFC, +ICF, +IPR — DTE Interface

+IFC is the flow control each way (DC1/DC3 in V.80's Synchronous Access Mode, Section 8.1), +ICF the DTE's character framing, +IPR the DTE's rate (0 automatic).

Table 38. DTE Interface Parameters

Name	Default	Range
+IFC	2,2	(0-3),(0-2)
+ICF	3,3	(1-6),(0-3)
+IPR	0	0, 300, 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600

&H*n* — Flow Control

The modem keeps the value in S62. Flow control is the pty's, so it changes nothing but what #UD reports.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

Table 39. &H*n* Commands

Command	Function (as reported)
&H0	None (*default*).
&H1	RTS/CTS.
&H2	XON/XOFF.
&H3	Both.

&I*n* — XON/XOFF Flow Control

USRobotics' software flow control: accepted, changing nothing.

Result code	When
`OK`	If *n* = 0–5.
`ERROR`	Otherwise.

\Q*n* — Flow Control

Accepted, changing nothing: flow control is the pty's.

Result code	When
`OK`	If *n* = 0–3.
`ERROR`	Otherwise.

\G*n* — XON/XOFF Pass Through

Accepted, changing nothing.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

&C*n* — Data Carrier Detect (DCD) Control

The modem keeps the value in S60. It changes nothing: a pty has no carrier line.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 40. &C*n* Commands

Command	Function
&C0	DCD always on (*default*).
&C1	DCD follows the carrier (on other modems).

`\Rn` — Ring Indicator After Answer

Accepted, changing nothing.

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

`\Tn` — Inactivity Timer

Accepted, changing nothing.

Result code	When
`OK`	If *n* = 0–255.
`ERROR`	Otherwise.

`+ILRR, +IDSR, +ILSD` — DTE Rate Report, DSR, Long Space

The DTE's rate reported, DSR, long space disconnect: kept. Each 0 or 1, *default* 0.

`+CRC` — Cellular Result Codes

27.007's: with `1`, each ring of a call is `+CRING: SYNC` for a call on a clear 64 kbit/s channel (an ISDN data call: SIP's CLEARMODE, a PBX's digital bearer, an mISDN card's unrestricted digital call) and `+CRING: VOICE` for any other, instead of `RING` (in words whatever `V` says). A program that answers by itself (dmodem-capid, CAPI on a D-Modem) tells the two apart by it. `&V` lists it.

Syntax:

- `+CRC=n`
- `+CRC? (gives +CRC: 0)`
- `+CRC=? (gives +CRC: (0,1))`

Result code	When
`OK`	If *n* = 0 (`RING`, the *default*) or 1.
`ERROR`	Otherwise.

---

12 Result Codes

Each command line is answered with a result code, as a word (`V1`) or a number (`V0`), each on a line of its own; `Q1` turns them off.

Table 43. Result Codes

Result code	Number	Description
`OK`	0	The command line ran.
`CONNECT`	1	The call is up (`X0`, fax and voice), or a fax command's data begins.
`CONNECT` *rate*	`Below`	A data call is up at *rate*.
`RING`	2	A call rings (`dmodemd`: every 3 s, Section 20).
`NO CARRIER`	3	No call came up, or the call is over.
`ERROR`	4	A command failed, or a line was too long.
`NO DIALTONE`	6	No dial tone (`X2`, `X4`).
`BUSY`	7	A busy tone (`X3`, `X4`).
`NO ANSWER`	8	No answer within `S7` seconds of dialing.
`FAX`	88	Adaptive answer (`+FAA=1`): a fax call; `Class 1` follows.
`DATA`	89	Adaptive answer: a data call; its `CONNECT` follows.

A data call's `CONNECT` gives the rate the modem receives at, in bit/s (`CONNECT 33600`), and with `V0` that rate's number (23). With `X0`, in fax and in voice mode it is `CONNECT` (1) alone. `\V1` adds the error correction and the compression (`CONNECT 33600/LAPM/V42bis`), and `&A` adds lines on the connection before it. Each rate's number is in the table below; a rate not in it is 1.

Table 44. CONNECT Rates and Their Numbers

Rate	n	Rate	n	Rate	n	Rate	n
75	80	24000	19	37333	51	50666	58
300	9	25333	81	38000	43	52000	32
600	10	26400	20	38666	52	53333	59
1200	5	26666	82	40000	25	54000	33
2400	11	28000	46	41333	53	54666	60
4800	12	28800	21	42000	26	56000	34

Table 45. S-Register Summary *(continued)*

Reg.	Description	Range	Unit	Default
S45	`V.cvm`: renegotiations (read only).	0–255	–	0
S47	Leased line (`&L`).	0–2	–	0
S48	`V.13` simulated carrier.	`0, 1`	–	0
S56	Result codes and dial tone (`X`).	0–4	–	4
S60	Carrier detect (kept).	`0, 1`	–	0
S62	Flow control (kept).	0–3	–	0
S70	`CONNECT`'s lines (`&A`).	0–7	–	0
S71	`CONNECT`'s rate (kept).	`0, 1`	–	0
S92	Answer delay.	0–255	s	2
S103	Error correction (`\N`).	0–5	–	3
S104	Data compression (`%C`).	0–3	–	3
S105	MNP block size (`\A`).	0–3	–	3
S106	Protocol in `CONNECT` (`\V`).	0–2	–	0
S161	`Y` (kept).	0–3	–	0

13.1 S-Register Definitions

The range given is what the register means; `ATS` accepts any value from 0 to 255 for any register but S44 and S45, which are read only. `ATZ` sets the registers to their *defaults*, but for S18, S28, S47, S48 and S161, which keep their values.

S0—Auto-Answer Ring Number

The number of rings the modem counts before it answers a call by itself. 0 disables auto-answer: the modem then answers only with `ATA`. On a leased line (`&L`) a value above 0 makes the modem answer, with no ring, whenever it is idle.

- Range: 0–255
- *Default*: 0
- Units: rings

11 `ISDN` and Cellular

`+B2` — B-Channel Protocol

D-Modem's own, with `CAPI` 2.0's numbers for layer 2: what runs on a clear 64 kbit/s channel (`+MS=CLEAR`, or a line that says its call is clear); other modes do not look at it. `&V` lists it.

Syntax:

- `+B2=`*n*
- `+B2?` (gives `+B2: 0`)
- `+B2=?` (gives `+B2: (0,5,7,9)`)

Result code	When
`OK`	If *n* = 0, 5, 7 or 9.
`ERROR`	Otherwise.

Table 41. +B2 Values

Value	Protocol
0	`X.75` (`LAPB` in `HDLC`, basic mode, a window of 7, the caller at address B), the protocol of `ISDN` terminal adapters, BBSes and mailers: the `DTE`'s characters go in I frames of up to 256 octets; `CONNECT 64000`, with `\V1` `CONNECT 64000/X.75`; `ATH` sends `DISC` (*default*).
5	Synchronous `PPP` (`RFC` 1618): the `DTE` speaks `PPP`'s asynchronous framing (`RFC` 1662, as pppd does on a serial port), each frame goes to the line in `HDLC` with its escapes undone and comes back escaped; `/PPP` with `\V1`.
7	The modem's link (`V.42`, `MNP` or plain characters, as `\N` says), for the clear channel of an earlier D-Modem.
9	`V.120` (`Q.921`'s frames for logical link 256 in extended mode, each I frame's information after `V.120`'s header octet `0x83`), the protocol of North American terminal adapters; `/V.120` with `\V1`.

`+CBST` — Bearer Service Type (V.110)

27.007's bearer service: `V.110`'s rate adaption on a clear channel (`+MS=CLEAR`). *name* 0 (the *default*) is asynchronous: the `DTE`'s characters through the modem's link (`\N`) whatever `+B2` says; 1 is synchronous: `+B2`'s protocol in `HDLC` at the user rate. *ce* has no use here. `CONNECT` gives the user rate (with `\V1` and no `V.42`, `/V.110`). `&V` lists it. See `docs/isdn.md`.

Syntax:

- `+CBST=[`*speed*`][,[`*name*`][,[`*ce*`]]]`
- `+CBST?` (gives `+CBST: 0,0,0`)
- `+CBST=?`

Result code	When
`OK`	For a *speed* below, *name* 0 or 1, *ce* 0 or 1.
`ERROR`	Otherwise.

Table 42. +CBST Speeds

Value	User rate
0	None (*default*): an answerer adopts the rate of the caller's frames (600 to 38400) if they come.
65, 66, 68, 70	300, 1200, 2400, 4800 bit/s (intermediate rate 8k).
71	9600 bit/s (16k).
75, 79	14400, 19200 bit/s (32k).
80, 81	28800, 38400 bit/s (64k).
83	56000 bit/s (7 bits an octet, no frames).

Table 44. CONNECT Rates and Their Numbers *(continued)*

Rate	n	Rate	n	Rate	n	Rate	n
7200	13	29333	47	42666	54	57333	83
9600	27	30666	48	44000	28	57600	35
12000	14	31200	22	45333	55	58000	44
14400	15	32000	40	46000	29	58666	84
16800	16	33333	49	46666	56	60000	45
19200	17	33600	23	48000	30	61333	85
21600	18	34666	50	49333	57	62666	86
		36000	42	50000	31	64000	87

13 S-Registers

The S-registers hold the modem's settings, 8 bits each. `ATS` reads and sets them: `ATS`*n*`?` gives register *n*'s value, `ATS`*n*`=`*v* sets it. Registers 0 to 255 are there; those not listed are 0 and free, kept as written. "Kept" below means set and read back, with no other effect.

Table 45. S-Register Summary

Reg.	Description	Range	Unit	Default
`S0`	Rings to answer after.	0–255	rings	0
`S1`	Ring counter.	0–255	rings	0
`S2`	Escape character.	0–255	`ASCII`	43
`S3`	Command line end.	0–255	`ASCII`	13
`S4`	Line feed.	0–255	`ASCII`	10
`S5`	Backspace.	0–255	`ASCII`	8
`S6`	Wait for dial tone.	0–255	s	2
`S7`	Wait for the call to come up.	0–255	s	60
`S8`	Comma's pause.	0–255	s	2
`S9`	Carrier detect time (kept).	0–255	0.1 s	6
`S10`	Carrier loss before hanging up.	0–255	0.1 s	7
`S11`	`DTMF` tone and gap.	0–255	ms	100
`S12`	Escape guard time (kept).	0–255	20 ms	50
`S13`	Echo (`E`).	`0, 1`	–	1
`S14`	Quiet (`Q`).	`0, 1`	–	0
`S15`	Verbose (`V`).	`0, 1`	–	1
`S16`	Tone or pulse dialing.	`0, 1`	–	1
`S18`	Loop test time.	0–255	s	0
`S24`	Hook flash time.	0–255	10 ms	20
`S28`	Pulse ratio (kept).	0–3	–	0
`S29`	Speaker control (`M`).	0–3	–	1
`S30`	Speaker volume (`L`).	0–3	–	3
`S31`	Automode (`N`).	`0, 1`	–	1
`S32`	Modulation (`+MS`).	`Table 22`	–	92
`S39`	Line quality (kept).	0–2	–	0
`S44`	`V.cvm`: tones loaded (read only).	0–255	tones	0

14 Fax Commands

With `AT+FCLASS=1`, the modem accepts `T.31`'s `Class 1` commands, the fax machine's side being the `DTE`'s (the `PBX` modules and libdmodem have a `T.30` engine of their own as well). As the caller, the modem sends `CNG` after dialing until the answerer's flags come in its first `+FRH=3` (`T.30` has it until the `DIS`).

`+FCLASS` — Service Class

This command sets whether the modem is a data modem, a `Class 1`, 1.0, `Class 2` or 2.0 fax modem or a voice modem. `ATZ` sets 0.

Syntax:

- `+FCLASS=`*n*
- `+FCLASS? (gives the class)`
- `+FCLASS=? (gives 0,1,1.0,2,2.0,8)`

Result code	When
`OK`	If *n* = 0, 1, 1.0, 2, 2.0 or 8.
`ERROR`	Otherwise.

Table 46. +FCLASS Values

Value	Class
0	Data (*default*).
1	Fax `Class 1` (`T.31`).
1.0	Fax `Class 1.0`: `Class 1` with `V.34` half-duplex fax (`T.31` Annex B, `+F34`).
2	Fax `Class 2` (Section 14.1).
2.0	Fax `Class 2.0` (`T.32`, Section 14.1).
8	Voice (`V.253`, Section 18).

S1—Ring Counter

The rings counted so far: `S0` compares it with its own value.

- Range: 0–255
- *Default*: 0
- Units: rings

S2—Escape Character

The `ASCII` value of the character of the escape sequence, `+++` (Section 19).

- Range: 0–255
- *Default*: 43 (+)
- Units: `ASCII`

S3—Command Line Termination Character

The character that ends a command line and the result codes.

- Range: 0–255
- *Default*: 13 (carriage return)
- Units: `ASCII`

S4—Response Formatting Character

The line feed after the result codes; a command line ends at it as well.

- Range: 0–255
- *Default*: 10 (line feed)
- Units: `ASCII`

S5—Command Line Editing Character

The backspace: it erases the last character typed. With echo on, the modem echoes it, a space and the backspace again.

- Range: 0–255
- *Default*: 8 (backspace)
- Units: `ASCII`

S6—Wait Time for Dial Tone

With `X2` or `X4`, how long the modem waits for a dial tone before `NO DIALTONE`: 2 s at least whatever the value, and 2 s more while a dial tone in a cadence is heard.

- Range: 0–255
- *Default*: 2
- Units: seconds

S24—Hook Flash Time

The length of a hook flash. The country sets it.

- Range: 0–255
- *Default*: 20 (the country's)
- Units: 0.01 s

S28—Pulse Dial Ratio

`&P`'s setting, kept: the country's table sets the pulse times. `ATZ` keeps it.

- Range: 0–3
- *Default*: 0
- Units: none

S29—Speaker Control

`M`'s setting.

- Range: 0–3
- *Default*: 1
- Units: none

S30—Speaker Volume

`L`'s setting.

- Range: 0–3
- *Default*: 3
- Units: none

S31—Automode

`N`'s setting, and `+MS`'s second parameter.

- Range: 0, 1
- *Default*: 1
- Units: none

S32—Modulation

`+MS`'s modulation, by its number (Table 22).

- Range: the numbers of Table 22
- *Default*: 92 (200 at 48000 Hz and more)
- Units: none

S39—Line Quality Monitor

`%E`'s setting, kept.

- Range: 0–2
- *Default*: 0
- Units: none

S44—V.cvm Tones Loaded

The tones loaded in the receive direction of the last `V.cvm` call (`+CVM`). Read only.

- Range: 0–255
- *Default*: 0
- Units: tones

S45—V.cvm Renegotiations

The renegotiations of the last `V.cvm` call. Read only.

- Range: 0–255
- *Default*: 0
- Units: none

S47—Leased Line

`&L`'s setting: 0 a dialled line, 1 a 2-wire leased line, 2 a 4-wire one. `ATZ` keeps it.

- Range: 0–2
- *Default*: 0
- Units: none

S48—Simulated Carrier Control

1: `V.13`'s simulated carrier control on a leased line, on a call with no error correction (`&L`). `ATZ` keeps it.

- Range: 0, 1
- *Default*: 0
- Units: none

S56—Result Code Set

`X`'s setting.

- Range: 0–4
- *Default*: 4
- Units: none

S7—Connection Completion Time-Out

How long the modem waits for the call to come up after dialing or answering: the answer and the training. The dial string's `W` and `@` wait this long too, and voice mode's `+VRN` no longer. `V.cvm`'s start-up takes 15 to 21 s, so `S7` should allow for it.

- Range: 0–255
- *Default*: 60
- Units: seconds

S8—Pause Time for Comma Dial Modifier

The pause of a `,` in the dial string. The country (`+GCI`) sets it.

- Range: 0–255
- *Default*: 2 (the country's)
- Units: seconds

S9—Carrier Detect Response Time

Kept: the modulations detect the carrier by themselves.

- Range: 0–255
- *Default*: 6
- Units: 0.1 s

S10—Automatic Disconnect Delay

How long the carrier may be gone before the modem hangs up with `NO CARRIER`.

- Range: 0–255
- *Default*: 7
- Units: 0.1 s

S11—`DTMF` Dialing Speed

The length of each `DTMF` digit dialed, and of the gap after it. A value below 40 dials at 95.

- Range: 0–255
- *Default*: 100
- Units: milliseconds

S12—Escape Guard Time

Kept: the escape's guard time is always 0.5 s.

- Range: 0–255
- *Default*: 50
- Units: 0.02 s

S13—Echo

`E`'s setting.

- Range: 0, 1
- *Default*: 1
- Units: none

S14—Quiet

`Q`'s setting.

- Range: 0, 1
- *Default*: 0
- Units: none

S15—Verbose

`V`'s setting.

- Range: 0, 1
- *Default*: 1
- Units: none

S16—Tone or Pulse Dialing

1 tone (`T`), 0 pulse (`P`).

- Range: 0, 1
- *Default*: 1
- Units: none

S18—Loop Test Timer

A loop test (`&T`) ends after this many seconds; 0, never. `ATZ` keeps it.

- Range: 0–255
- *Default*: 0
- Units: seconds

S60—Data Carrier Detect

`&C`'s setting, kept.

- Range: 0, 1
- *Default*: 0
- Units: none

S62—Flow Control

`&H`'s setting, kept.

- Range: 0–3
- *Default*: 0
- Units: none

S70—Connection Lines

`&A`'s setting.

- Range: 0–7
- *Default*: 0
- Units: bits

S71—Rate in CONNECT

`&E`'s setting, kept.

- Range: 0, 1
- *Default*: 0
- Units: none

S92—Answer Delay

How long the modem waits after going off hook to answer before its answer tone; 0 waits 10 ms. Elsewhere `S92` is a fax transmit level: `ATS92=10` here makes the modem wait 10 s.

- Range: 0–255
- *Default*: 2
- Units: seconds

S103—Error Correction

`\N`'s setting; setting the register is `\N`.

- Range: 0–5
- *Default*: 3
- Units: none

S104—Data Compression

`%C`'s setting, and `&K`'s.

- Range: 0–3
- *Default*: 3
- Units: none

S105—`MNP` Block Size

`\A`'s setting.

- Range: 0–3
- *Default*: 3
- Units: none

S106—Protocol in CONNECT

`\V`'s setting.

- Range: 0–2
- *Default*: 0
- Units: none

S161—Long Space Disconnect

`Y`'s setting, kept. `ATZ` keeps it.

- Range: 0–3
- *Default*: 0
- Units: none

+FAA — Adaptive Answer

With +FAA=1, a call the modem answers (ATA, S0) in data or fax mode is answered as a data call (its answer tone, V.8's ANSam, serves a fax machine as its CED), and found out to be a fax machine's or a data modem's:

- a fax call by the caller's CNG (1100 Hz, 0.5 s of every 3.5 s; or a burst cut short as the answer tone starts, a fax machine that stops its CNG on hearing it), the V.8 CM of a fax call (a call function of facsimile, as V.34 fax machines and some Class 1 modems send it on ANSam), the far end going over to T.38 (dmodemd's SIP calls), or a caller silent through every answer signal of the automode chain (ANSam, then V.32's, V.22's, V.21's, V.23's and Bell 103's, some 20 s, within T.30's T1 of 35 s): `FAX`, +FCLASS becomes 1, and the modem goes on as a Class 1 answerer: what is left of CED's 3 s, then `CONNECT`, as after AT+FTH=3, for the DTE's first frames;
- a data call by its modem's handshake: `DATA` when it comes up, then its `CONNECT`; +FCLASS becomes 0.

A call that ends with neither leaves +FCLASS as it was. A V.22, V.21 or Bell 103 caller, silent until it hears its own answer signal, answers its stage of the chain: `DATA`. In voice mode see Section 18: a calling tone on the call switches it to fax or data. Rockwell's modems give `FAX` and `DATA` too, as 33 and 35 with V0: here those are rates' CONNECTs, so 88 and 89.

Syntax:

- +FAA=*n*
- +FAA? (gives `+FAA: 0`)
- +FAA=? (gives `+FAA: (0,1)`)

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

Table 47. +FAA Values

Value	Meaning
0	Answer as +FCLASS says (*default*; Z sets it).
1	Adaptive answer: fax or data.

+FPR — Fax DTE Rate

The fax DTE's rate: kept, with no effect. 0 (*default*), 1, 2, 4, 8, 10 or 18.

+F34 — V.34 Fax Rates

In Class 1.0 (+FCLASS=1.0), a fax call starts with V.8 and V.34 half-duplex (T.30 Annex F) instead of CNG or CED, as fax software such as HylaFAX expects with AT+F34. When both ends have it, the modem gives `+F34:p,c` (the primary channel's rate in 2400 bit/s, the control channel's: 1 for 1200 bit/s), then `CONNECT`, and from then on the call is T.31's V.34 data mode, with no more commands:

- the DTE's frames, without their FCS, each ended by <DLE><ETX>, and the modem's, with their FCS, the same way;
- <DLE><0x6B> from the DTE (the sender) asks for the primary channel; the modem gives <DLE><0x6B><DLE><r> when it is up (*r* = 0x6F + the rate in 2400 bit/s), at either end, and the ECM block's frames follow;
- <DLE><0x6D> from the DTE ends the block; the modem gives <DLE><0x6D><DLE><0x6E> when the control channel (1200 bit/s) is back, at either end;
- <DLE><EOT> from the DTE ends the call (`OK`); the modem gives it when the line goes (`NO CARRIER`).

With a fax machine that has no V.8, the call goes on as Class 1: the caller's first `CONNECT` is for the answerer's frames, as after +FRH=3; the answerer's is for its own, as after +FTH=3 (its ANSam was the CED). V.34 fax needs ECM, and the modem does not poll on it. Not on T.38, nor after adaptive answer. ATZ sets the defaults.

Syntax:

- +F34=[*maxp*][,[*minp*][,[*prefc*][,[*maxp2*][,*minp2*]]]]
- +F34? (gives `14,1,0,14,1`)
- +F34=? (gives `(0-14),(0-14),(0-2),(0-14),(0-14)`)

Result code	When
`OK`	If each value is in its range.
`ERROR`	Otherwise.

Table 49. +F34 Values

Value	Meaning
maxp	The primary channel's highest rate, in 2400 bit/s (14: 33600, *default*); 0: no V.34, Class 1 as before.

Table 49. +F34 Values *(continued)*

Value	Meaning
minp	Its lowest (1: 2400, *default*).
prefc	The control channel's rate wanted (0 any, *default*; 1 1200; 2 2400): kept, the modem's is 1200.
maxp2, *minp2*	For the other direction: kept.

14.1 Class 2 and 2.0

With AT+FCLASS=2.0 (T.32) or AT+FCLASS=2 (the older Class 2 that fax programs know), the modem runs T.30 itself: the DTE sends and receives the pages in T.4 MH with EOLs, <DLE> doubled, ended by <DLE><ETX>, and steers the call with the commands below. Class 2 has the older names: +FLID, +FDCC, +FDIS, +FDCS, +FBOR, +FK, +FPTS and +FHNG (either name works in either class), and its reports say `+FCON`, `+FCSI`, `+FTSI`, `+FDIS`, `+FDCS`, `+FPTS` and `+FHNG`. The rates are V.17, V.29 and V.27ter's; error correction mode, when +FIS has it, is the modem's own (the DTE's data is the same). The pages are MH whatever the remote uses (MR or ECM: the modem codes them again), A4 width. A call goes over to T.38 as a Class 1 call does (the same commands and results). +FAA (adaptive answer) is Class 1's, and Class 1's +FTH, +FRH, +FTM, +FRM, +FTS and +FRS are `ERROR` in Class 2 mode.

Example session: sending two pages, then receiving them

```
ATD<number>       +FCO           the DIS came
                  +FCI:"<id>"
                  +FIS:1,5,0,2,0,0,0,0
                  OK
AT+FDT            +FCS:1,5,0,2,0,0,0,0
                  CONNECT        DCS, training, CFR
<page><DLE><ETX>  OK
AT+FET=0          +FPS:1,0,0,0,0
                  OK             MPS, MCF
AT+FDT            CONNECT
<page><DLE><ETX>  OK
AT+FET=2          +FPS:1,0,0,0,0
                  +FHS:00
                  OK             EOP, MCF, DCN

ATA               +FCO           the CFR went out
                  +FTI:"<id>"
                  +FCS:1,5,0,2,0,0,0,0
```

+FCC, +FIS, +FCS — Capabilities, Session Parameters

The capabilities (+FCC, Class 2's +FDCC, which sets +FIS as well) and the session's (+FIS, Class 2's +FDIS, read again at each +FDT). +FCS? gives the last session's. At the start 1,5,0,2,0,0,0,0.

Syntax:

- +FCC=*vr*,*br*,*wd*,*ln*,*df*,*ec*,*bf*,*st*
- +FIS=*vr*,*br*,...
- +FCC?, +FIS?, +FCS?
- +FCC=?, +FIS=? (give (0,1),(0-13),(0),(0-2),(0),(0-2),(0),(0-7))

Result code	When
`OK`	If every value is in its range.
`ERROR`	Otherwise.

Table 51. +FCC Values

Value	Meaning
vr	0 normal, 1 fine resolution.
br	0 to 5: 2400, 4800, 7200, 9600, 12000, 14400 bit/s. 6 to 13: V.34 half-duplex (T.30 Annex F) as well, 16800 to 33600 bit/s, with error correction mode only: V.8 offers it at the start of the call, and a remote without it has V.17 and the rest. A caller that polls sets +FSP=1 before dialling, for V.8 to say so. +FCS then gives the V.34 rate.
ec	0 none, 1 or 2 error correction mode.
others	Taken in their ranges.

`+FMI`, `+FMM`, `+FMR` — Fax Identification

These commands give the manufacturer, the model and the revision, as `+GMI`, `+GMM` and `+GMR` give them, in any mode.

Result code	When
`OK`	Always.

`+FTS`, `+FRS` — Transmit Silence, Receive Silence

`+FTS` sends silence for *n* times 10 ms; `+FRS` waits for silence of *n* times 10 ms.

Syntax:

- `+FTS=`*n*
- `+FRS=`*n*

Result code	When
`OK`	When it is done.
`ERROR`	Outside fax mode, or with no *n*.

`+FTM`, `+FRM` — Transmit Data, Receive Data

These commands send or receive data at modulation *m*: `CONNECT`, then the data with <DLE> doubled, ended by <DLE><ETX>.

Syntax:

- `+FTM=`*m*`, +FRM=`*m*
- `+FTM=?, +FRM=?  (give 24,48,72,73,74,96,97,98,121,122,145,146)`

Result code	When
`CONNECT`	Then `OK` after the data.
`ERROR`	Outside fax mode, or for a modulation the modem has not.

Table 48. +FTM and +FRM Modulations

m	Modulation
24, 48	V.27ter, 2400 and 4800 bit/s.
72, 96	V.29, 7200 and 9600 bit/s.
73, 97, 121, 145	V.17, 7200 to 14400 bit/s, long training.
74, 98, 122, 146	V.17, 7200 to 14400 bit/s, short training.

`+FTH`, `+FRH` — Transmit `HDLC`, Receive `HDLC`

These commands send or receive `HDLC` frames at `V.21` channel 2 (300 bit/s), the only modulation for them.

Syntax:

- `+FTH=3, +FRH=3`
- `+FTH=?, +FRH=?  (give 3)`

Result code	When
`CONNECT`	Then `OK` or `ERROR` after each frame.
`ERROR`	Outside fax mode.

```
                    OK
AT+FDR              CONNECT
<DC2>               <page><DLE><ETX>
                    +FPS:1,<lines>,0,0,0
                    +FET:0
                    OK
AT+FDR              ... +FET:2, OK
AT+FDR              +FHS:00
                    OK
```

In `Class 2` an `XON` follows the sending `CONNECT`. `+FDR`'s page waits for the `DTE`'s <DC2> (a second at most); the pages are accepted and confirmed as they come, and wait for `+FDR`. When the call ends, the modem reports the hangup status and the result of the command then under way (`NO CARRIER` for `ATD` and `ATA`), and goes on hook.

Table 50. +FHS Hangup Status

Value	Meaning
00	The call was done.
02	`+FKS` ended it.
11	No fax machine answered.
2x	Sending, phase B (the `DIS`, the training).
4x, 5x	Sending a page, or the post-page exchange.
7x, 9x	Receiving.

`Class 2.0` gives the status in hexadecimal digits (`T.32` Table 23), `Class 2` in decimal.

`+FLI`, `+FPI` — Local ID, Polling ID

Our identity in `TSI` and `CSI` (`+FLI`, `Class 2`'s `+FLID`) and in `CIG` when we poll (`+FPI`, `Class 2`'s `+FCIG`; empty: `+FLI`'s): up to 20 printable characters, the blanks in the quotes kept.

Syntax:

- `+FLI="`*id*`", for example AT+FLI="Fax ID with Spaces"`
- `+FPI="`*id*`"`
- `+FLI?, +FPI? (give the identity in quotes)`
- `+FLI=?, +FPI=?  (give (20)(32-127))`

Result code	When
`OK`	Always.
`ERROR`	For more than 20 characters.

`+FLP`, `+FSP` — Document to Poll, Request to Poll

`+FLP=1` (`Class 2`'s `+FSPL`): we have a document to be polled; the `DIS` says so, and a caller's `DTC` has `ATA` report `+FCO`, `+FPI:"`*cig*`"` and `+FTC:`*dtc* (`Class 2`: `+FCIG`, `+FDTC`) and `OK`, for `+FDT` to send it. `+FSP=1`: we poll; after a `DIS` that offers a document (`ATD` reports `+FPO`, `Class 2 +FPOLL`), `+FDR` sends the `CIG` and `DTC` and receives as the answerer does (`+FTI`, `+FCS`, `CONNECT` and the page).

Syntax:

- `+FLP=`*n*`, +FSP=`*n*
- `+FLP?, +FSP?`
- `+FLP=?, +FSP=?  (give (0,1))`

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise; the caller's `+FDR` is `ERROR` without `+FSP=1`.

18 Voice Commands

With `AT+FCLASS=8` the modem accepts `V.253`'s voice commands; they are `ERROR` in the other modes, but for `+VCID`. Each `+V` parameter accepts `?` (its values) and `=?` (their ranges).

`+VIP` — Initialize Voice Parameters

This command sets the voice settings as at the start.

Result code	When
`OK`	Always.

`+VNH` — Automatic Hangup Control

0 or 1. It is accepted and has no effect: a held call's failed handshake hangs up.

`+VGT`, `+VGR` — Transmit and Receive Gain

The gain sending (`+VGT`) and receiving (`+VGR`), 0 to 128.

Result code	When
`OK`	If n = 0–128.
`ERROR`	Otherwise.

`+VIT` — Inactivity Timer

The inactivity timer, 0 to 255.

`+VSD` — Silence Detection

Silence detection's sensitivity and period, each 0 to 255.

Syntax:

- `+VSD=`*s*`,`*t*

`+FBO`, `+FCR`, `+FPS`, `+FHS` — Bit Order, Receive, Page Status, Hangup Status

`+FBO` (`Class 2`'s `+FBOR`): bit 0 set, the page data's first bit is the top of each octet (0, the *default*: its lowest). `+FCR` and `+FPS` are kept. `+FHS?` gives the last hangup status. Other `+F` settings that fax programs send in `Class 2` (`+FNR`, `+FIE`, `+FCT` and the like) are kept with no effect.

Syntax:

- `+FBO=`*n*`, +FCR=`*n*`, +FPS=`*n*
- `+FHS?`

Result code	When
`OK`	Always.

`+FDT`, `+FDR`, `+FET`, `+FKS` — Send, Receive, End Page, End Call

`+FDT` sends a page, `+FDR` receives one, `+FET` gives the command after the page sent, and `+FKS` (`Class 2`'s `+FK`) ends the call with a `DCN` (`+FHS:02`), as in the example above.

Syntax:

- `+FDT, +FDR, +FKS`
- `+FET=`*n*

Result code	When
`CONNECT`	Then the page and `OK`; or the reports, then `OK`.
`ERROR`	With no `Class 2` call.

Table 52. +FET Values

Value	Meaning
0	`MPS`: another page follows.
1	`EOM`: another document follows.
2	`EOP`: the last page.

`+MV18AM`, `+MV18P` — Answer Message, Probe Order

`+MV18AM` sets the answer message, up to 63 printable characters, the blanks in its quotes kept (`AT+MV18AM="GA"`): sent in the probes, and on answering with `+MV18S`'s *msg*. `+MV18P` (2 to 7) is kept: the probes go in the order above.

Syntax:

- `+MV18AM="`*text*`", +MV18AM?, +MV18AM=?` (gives `+MV18AM: (63)(32-126)`)
- `+MV18P=`*n*`, +MV18P?` (gives `+MV18P: 2`), `+MV18P=?`

Result code	When
`OK`	Always.
`ERROR`	For a message too long or a value out of range.

`+MBAUDOT` — 5-bit Mode

This command (D-Modem's own) sets the 5-bit mode's rate, tones and figures, for telex and `RTTY`: `AT+MBAUDOT=50,2125,2295,1` is 50 baud ITA2 on `RTTY`'s 170 Hz shift. The two ends must agree; automoding listens on these tones. `Z` sets the US TTY's `45,1400,1800,0`.

Syntax:

- `+MBAUDOT=[`*rate*`][,[`*mark*`][,[`*space*`][,[`*code*`]]]]`
- `+MBAUDOT?` (gives `+MBAUDOT: 45,1400,1800,0`)
- `+MBAUDOT=?` (gives `+MBAUDOT: (45,50,75,100),(300-3400),(300-3400),(0,1)`)

Result code	When
`OK`	Always.
`ERROR`	For a value out of range, or the same mark and space.

Table 54. +MBAUDOT Values

Value	Meaning
rate	45 (45.45), 50, 75 or 100 bit/s.
mark	The mark (1) tone, Hz.
space	The space (0) tone, Hz.
code	The figures: 0 the US TTY's ($ on D, " on Z, ; on V, unshift on space), 1 ITA2's (telex).

15 Text Telephone Commands

The modem is a text telephone in `V.18`'s modes (+MS or +MV18S chooses one): the 5-bit (Baudot) mode of the TTYs, at 45.45 bit/s on 1400 and 1800 Hz (TIA-825-A), and with +MBAUDOT the telex and RTTY variants (50 bit/s, ITA2's figures, other tones); EDT (`V.21`'s channel 1 at 110 bit/s, 7 bits, even parity, 2 stop bits); and `V.21`, `V.23` and Bell 103 as their data modes. The 5-bit mode and EDT are half duplex: the modem sends 150 ms of mark, the characters, and 300 ms of mark after the last, and does not hear the line meanwhile; it starts no characters while the remote's carrier is on. Neither hangs up for want of carrier. The DTE sends and receives ASCII: in the 5-bit mode lower case goes as capitals, the shift codes are the modem's, and characters the code has not are dropped. A text telephone call has no answer tone: dialling, the modem goes into its mode as the dialling ends; answering, at once.

AT+MS=V18 (or AT+MV18S=1) finds the remote's mode. It listens for 3 s, for a 5-bit character, an EDT character, `V.21`'s channel 1 at 300 bit/s (or a steady mark), a Bell 103 or `V.23` caller's mark (answering), or `V.21`'s channel 2, Bell 103's 2225 Hz or `V.23`'s 1300 Hz (calling). Hearing nothing it probes, in turn: the answer message (+MV18AM) and a new line in the 5-bit mode, then in EDT, each followed by 2 s of listening, then 3 s each of its own `V.21`, Bell 103 and `V.23` mark, listening for the answer to it; and over again until S7 runs out. Found, a 5-bit or EDT remote is answered in its mode (once it is quiet, so that another automoding modem, whose probe this may have been, finds it too); `V.21`, `V.23` and Bell 103 go on as those modems do. DTMF text telephones (`V.18`'s DTMF mode) are not supported.

+MV18S — V.18 Selection

This command chooses the text telephone mode (as +MS would; 0 sets +MS=92 if +MS had a text one), and whether the modem probes a silent remote and sends the answer message on answering. Z sets `0,0,0,0,1`.

Syntax:

- `+MV18S=[mode][,[dflt][,[fbk][,[msg][,[probe]]]]]`
- `+MV18S? (gives +MV18S: 0,0,0,0,1)`
- `+MV18S=? (gives +MV18S: (0-2,4-7),(0-7),(0,1),(0,1),(0-2))`

Result code	When
`OK`	Always.
`ERROR`	For a value out of range, or *mode* 3 (DTMF).

Table 53. +MV18S Values

Value	Meaning
mode	0 off; 1 automoding (+MS=18); 2 5-bit (45); 4 EDT (110); 5 `V.21`; 6 `V.23`; 7 Bell 103.
dflt	The *default* answering mode: kept.
fbk	Kept.
msg	1: the answer message on answering in the 5-bit mode or EDT.
probe	0: automoding only listens; 1 or 2: it probes.

+MV18R — V.18 Report

With +MV18R=1 automoding reports the mode it found before the `CONNECT`, as `+MV18R: mode` with +MV18S's numbers (`+MV18R: 2` for a TTY).

Syntax:

- `+MV18R=n`
- `+MV18R? (gives +MV18R: 0)`
- `+MV18R=? (gives +MV18R: (0,1))`

Result code	When
`OK`	If *n* = 0 or 1.
`ERROR`	Otherwise.

---

16 Alarm Reports: Contact ID

With AT+MS=CONTACTID the modem speaks Contact ID (SIA DC-05), the DTMF format in which alarm panels report events to a central station's receiver. There are no commands of its own: ATA (or S0) makes the modem the receiver, ATD the panel, and the call is in data mode from the handshake on, `CONNECT 40` on both sides. Neither hangs up for want of carrier; ATH or the far end ends the call.

A message is 16 digits: the account (4), the message type (18, or 98), the qualifier (1 a new event or an opening, 3 a restore or a closing, 6 a previous event still there), the event code (3), the partition (2), the zone or user (3), and the checksum, which makes the digits' values add up to a multiple of 15 (0 counts 10; B to F are 11 to 15, sent as ∗, #, A, B and C).

The Receiver. Answering, the modem sends the handshake (1400 Hz, then 2300 Hz, 100 ms each), and again after 3 s with no message (four times in all). Each message with a good checksum goes to the DTE as a line of its 16 digits (`1234181131010158`) and gets the kissoff (1400 Hz for 900 ms); a bad one gets nothing, and the panel sends it again. What the DTE types is dropped.

The Panel. Dialling, the modem waits for the handshake, then sends each line the DTE types (`1234 18 1131 01 015`): 16 digits, or 15 and the modem adds the checksum; spaces and dashes are ignored. A message goes 250 ms after the handshake or the last kissoff, 55 ms a digit and 55 ms between, and again (four times in all) when no kissoff comes within 1.25 s. Each line gets `ACK` (kissed off), `NAK` (four times without), or `ERROR` (no message).

---

17 Text Messages: Fixed-Line SMS

With AT+MS=SMS the modem speaks fixed-line SMS (ETSI ES 201 912, protocol 1), the way some home telephones send and receive text messages through their telephone company's message centre. There are no commands of its own: ATA (or S0) makes the modem the receiving side, ATD the sending side, and the call is in data mode from the start, `CONNECT 1200` on both sides. The line is `V.23`'s 1200 bit/s channel (1300 and 2100 Hz) both ways, one side at a time.

Receiving. Answering, the modem invites the caller's messages, takes each one, and gives it to the DTE: a delivery (a message centre's to a telephone) as `+CMT: "from",,"time"`, a submission (a telephone's to its message centre) as `+SUBMIT: "to"`, each followed by a line of its text (UTF-8; 8-bit data in hex). Anything else comes as `+PDU: hex`. The call ends when the caller says it has no more; what the DTE types is dropped.

Example session: sending two pages, then receiving them

```
+CMT: "+441234567890",,"26/10/31,09:05:07+00"
Your code is 1234
```

Sending. Dialling, the modem waits for the invitation, then sends each line the DTE types:

Table 55. Lines a Sending Modem Takes

Line	Sends
number,*text*	A submission of *text* to *number* (a telephone's message).
<*number*,*text*	A delivery of *text* from *number* (a message centre's), stamped with the time now (UTC); a *number* with letters is a name.
:*hex*	The message (3GPP TS 23.040's TPDU) as it is.

A *number* is digits (and ∗, #), + first for an international one. The text goes in GSM's 7-bit alphabet, up to 160 characters, or in UCS2, up to 70, when it has a character that alphabet lacks. Each line gets `OK` (taken), `+CMS ERROR: cause` (refused, with the far end's cause), or `ERROR` (not a message, or no answer after three tries). With no line for 3 s, the modem says it has no more and hangs up.

`+VSM` — Voice Compression Method

This command sets the format of the samples `+VTX`, `+VRX` and `+VTR` carry: format *m* at *r* samples/s (left out: 8000, or the rate the format runs at). *s* and *e*, silence compression's, are accepted (0 to 255) and kept, though the modem compresses no silence.

Syntax:

- `+VSM=`*m*`[,`*r*`[,`*s*`[,`*e*`]]]`
- `+VSM? (gives +VSM=1,8000,0,0 at the start)`
- `+VSM=?  (lists the formats, one a line)`

Result code	When
`OK`	Always.
`ERROR`	For another format, or a rate the format does not run at; the setting is kept.

Table 56. +VSM Formats

m	Format
`0, 1`	8-bit linear, signed and unsigned (offset by 128), at 7200, 8000 or 11025 (`V.253`'s numbers).
`4, 5`	`G.711` u-law and A-law, at 8000.
128	8-bit linear unsigned, as 1 (Conexant's numbers from here on).
129	16-bit linear signed, the low octet first, at 7200, 8000 or 11025.
`130, 131`	A-law and u-law, as 5 and 4.
132	IMA `ADPCM`, 4 bits a sample, two an octet with the first in the low nibble (as in a `WAV` file), at 7200, 8000 or 11025; the coder starts afresh at each `+VTX`, `+VRX`, `+VTR`.

`+VTX, +VRX, +VTR` — Voice Transmit, Receive, Duplex

These commands send, receive, or both at once: `CONNECT`, then samples in `+VSM`'s format and rate, `<DLE>` doubled. A gap in the `DTE`'s samples plays as silence.

Result code	When
`CONNECT`	Then `OK` when it ends.
`ERROR`	Without the line or a handset taken (`+VLS`).

Table 58. Voice Data Commands

Command	Ends at
`+VTX`	`<DLE><ETX>`, `<DLE>!` or `<DLE><CAN>` from the `DTE`; then `OK`.
`+VRX`	Any character from the `DTE`; the modem ends its samples with `<DLE><ETX>`, then `OK`.
`+VTR`	`<DLE>^` from the `DTE` only; then `<DLE><ETX>` and `OK`.

`+VTD` — Beep Tone Duration

The duration of a `+VTS` element that gives none, in 10 ms, 0 to 6000 (0: 1 s); 100 at the start.

`+VTS` — `DTMF` and Tone Generation

This command plays the elements of *string*, separated by commas, in turn, and gives `OK` after the last. Durations are in 10 ms, 0 to 6000. A field may be left empty: a frequency is then 0, a digit none (silence), and a duration `+VTD`'s, so `{5}`, `[440]` and `[]` last `+VTD` and `{}` is silence for as long. Two digits in a row get 50 ms of silence between them: `+VTS=1,2,3` dials 123. A hook flash is the line's: over `SIP` an `RFC` 4733 event (16), the audio silent meanwhile.

Syntax:

- `+VTS=`*string*
- `+VTS? (gives the tone settings)`
- `+VTS=?  (gives +VTS=0-4000,0-4000,0-6000)`

Result code	When
`OK`	After the last element.
`ERROR`	If one element cannot be (nothing is played): a flash on a line that has none (a `SIP` call whose far end accepts no telephone-events), a value out of range.

Table 59. +VTS Elements

Element	Plays
s	The `DTMF` digit *s* (0–9, *, #, A–D), or `!` a hook flash, for `+VTD`.
`{`*s*`,`*d*`}`	The `DTMF` digit or hook flash *s* for *d*.
`[`*f1*`,`*f2*`,`*d*`]`	A tone of *f1* and *f2* Hz (0 to 4000; 0 is none) for *d*.

18.1 Voice Events

While the line is taken (`+VLS=1`) the modem reports what it hears, as `V.253` has it, as `<DLE>` and a character. `DTMF` digits from the far end are reported 51 ms into them and again every 70 ms while they last, in the voice command state as well as among `+VRX`'s samples, whose audio is silence from a digit's report until it ends; the modem's own `+VTS` tones are not reported. The calling tones are reported as a burst ends and 4 s apart at least.

19 Online, the Escape and Hanging Up

Online, what the `DTE` sends goes to the other modem, except the escape sequence: three `S2` characters (`+++`), each within 0.5 s of the one before, with 0.5 s of nothing before them and after them. The modem then says `OK` and is in online command mode, the call still up: `ATO` goes back online, `ATH` hangs up, and on a `V.92` connection `AT+PMHR` asks for modem-on-hold. The `+++` itself is data (the modem cannot know it is the escape until the guard time after it), and what the `DTE` sent before the escape and the modem had not sent yet is kept, not dropped: it goes to the other modem, `+++` last, once `ATO` is back online.

The call ends with `NO CARRIER` when the other side hangs up or its carrier is gone for `S10` tenths of a second. `dmodemd` hangs up as well when the `DTE` drops `DTR` (it sets the pty's speed to 0) or, with HUPCL set on the pty, closes it; it then makes a new pty (`/dev/ttyDM0` follows it).

+VLS — Analog Source/Destination Selection

This command takes the line for voice or leaves it, or puts the voice engine on a handset. Moving between 1, 14 and 15 keeps the voice engine running, the line taken or left.

Result code	When
`OK`	Always.
`ERROR`	For a device the modem has not.

Table 57. +VLS Values

Value	Device
0	Leave the line, on hook (*default*).
1	Take the line for voice (on `SIP`: answer the call that rings).
14	A handset of the modem's own alone, the line on hook: `+VRX` records its microphone, `+VTX` plays to its earpiece, and `DTMF` from it is reported.
15	The line with the handset on the call: the `DTE` hears the line and the handset, the line gets the `DTE`'s samples and the handset.

`AT+VLS=?` gives `V.253`'s list of the devices, with the events each reports in the transmit, receive and command states as hex bit fields (bit *n* for `V.253`'s event *n*, event 0 the leftmost of 32):

Example output: AT+VLS=?

```
0,"",00000000,00000000,90000000
1,"T",0B043060,0B043060,0B043060
```

On hook (0) that is caller ID (with `+VCID`) and `RING`; on the line (1) `DTMF`, the two calling tones, the loop current interruption, a ringing tone, a busy tone and the two answer tones, in every state. A line that senses the phone on the modem's own jack (a SmartUSB56 with `dmodemd -usb-handset`) adds event 8 to both, in every state (`00800000, 0B843060, ...`). A modem with a handset of its own (a SmartUSB56 with `dmodemd -usb-voice-codec`) lists two more:

Example output: AT+VLS=?, the handset's lines

```
14,"H",08000000,08000000,08000000
15,"HT",0B043060,0B043060,0B043060
```

+VEM — Event Reporting and Masking

0 or 1: `V.253`'s event reports in the speakerphone state, which the modem has not. Kept; 0 at the start and with `+VIP`.

D — Voice Dial

In voice mode, `ATD`*number* dials and gives `OK` when the far end answers, `BUSY`, NO ANSWER after `S7`, or `ERROR` when the call fails otherwise. An answer tone is the answer and a busy tone `BUSY` on any line (on `SIP` in its early media too); on `SIP` the answer is otherwise the 200 `OK`, on a line that does not tell the ringing tone's end (`+VRA`), or `+VRN` without one. On an analogue line the modem waits 2 s for the dial tone first. With the line taken it dials the digits as `DTMF`, `S11` ms each.

H — Voice Hang Up

In voice mode, `ATH` leaves the line and goes back to data mode, `+FCLASS=0`.

+VRA — Ringback Goes Away Timer

The far end has answered when the ringing tone has been gone this long since its last ring, in 100 ms, 0 to 255 (0: as the first ring ends); 50 at the start.

+VRN — Ringback Never Appeared Timer

The far end has answered when no ring has come in this long after dialling, in seconds, 0 to 255 (not beyond `S7`); 10 at the start.

Table 60. Voice Events

Report	Event
`<DLE>0-9 * # A-D`	A `DTMF` digit from the far end.
`<DLE>~`	The `DTMF` digit ended.
`<DLE>r`	A ring of a ringing tone (400 to 480 Hz, or 440 and 480 Hz, or 400 and 450 Hz, on for 0.6 s at least; the UK's two bursts are one ring; it ends after 0.3 s without the tone).
`<DLE>b`	A busy or congestion tone (four rounds of 0.2 to 0.8 s on and off).
`<DLE>a`	An answer tone of 2100 Hz (a fax or data answer), 0.3 s into it.
`<DLE>f`	An answer tone of 2225 Hz (a data answer), 0.3 s into it.
`<DLE>c`	A calling fax machine's `CNG` (1100 Hz, bursts of 0.36 to 0.8 s).
`<DLE>e`	A calling data modem's `V.25` tone (1300 Hz).
`<DLE>l`	Loop current interruption: the far end ended the call.
`<DLE>H, <DLE>h`	The phone on the modem's own jack off hook, on hook again.
`<DLE>X ...<DLE>.`	Caller ID (`+VCID`).

When the far end ends a voice call, the modem reports `<DLE>l` (`V.253`'s loop current interruption, "usually a remote hangup") and goes on hook (`+VLS=0`); `V.253` has no `NO CARRIER` in voice mode. What was under way ends: `+VRX`'s samples with `<DLE><ETX>`, then `OK`; `+VTX` and `+VTR` when the `DTE` ends them (its samples until then are discarded), then `OK`, `+VTR`'s after `<DLE><ETX>`; `+VTS`'s tones with `ERROR`. In the command state `<DLE>l` comes alone.

18.2 From Voice to Fax or Data

A voice call can become a fax or data call (`V.253` 5.3.2 and its example `I.6`). `+FCLASS=1` (or `0`) in the voice command state, off hook, keeps the call: the voice engine stops, the line stays off hook and silent, and `A` answers on it as a fax (`Class 1`: `CED`, then `CONNECT` for the `DIS`) or data modem, or `D` calls on it (call progress waits for the far end's answer tone). `H`, `Z` or the far end hanging up ends the held call. For example, a fax machine calls while the greeting plays:

Example session: from voice to fax

```
AT+VTX                   CONNECT       the greeting's samples follow
                         <DLE>c        the caller's CNG
<DLE><ETX>               OK
AT+FCLASS=1              OK
ATA                      CONNECT       class 1, as after AT+FTH=3
```

With `+FAA=1` the modem does it itself, in the voice command state (a `+VTX`, `+VRX`, `+VTR` or `+VTS` under way ends first): after `<DLE>c`, `FAX`, `+FCLASS=1` and the fax answer's `CONNECT`; after `<DLE>e`, `+FCLASS=0` and the adaptive data answer (`DATA`, then its `CONNECT`).

20 `dmodemd` on a `SIP` Line

- `ATD`*number* calls `sip:`*number*`@`*domain* of `SIP_LOGIN`. A telephone number (only `V.250`'s dial characters: digits, `* # + A B C D`, the modifiers `T P W , @ ! ;` and `- ( ) .` and blanks) goes as its digits and `DTMF` characters: `ATDT555-1000` calls `sip:5551000@`*domain*. Anything else is a `SIP` user's name, as it is: `ATDalice`.
- The line has no dial tone: `ATX3` (or `X1`), else `NO DIALTONE`.
- A dial string with a `;` (`ATDT5551000;`) makes the `SIP` call and says `OK` at once, the modem in command state: what `ATD` adds then goes as `DTMF` on the call (a menu's choices, an extension), and `ATD` alone or `ATA` starts the modem's handshake.
- With `-L` `dmodemd` registers and accepts calls: `RING` every 3 seconds while one waits, `ATA` or `S0` rings (`ATS0=1`) answer it, and the rings stop if the caller gives up. A call that comes while the line is in use is refused (486, busy); without `-L` every call is (480).
- The call is `G.711` at 8000 Hz, which is what `V.90` and `V.92` need (`AT+MS=92`, the *default*). Over a real `PSTN` gateway a lower modulation (`AT+MS=132,0,4800,9600`) may do better.

Term	Meaning
`DCD`, `DSR`, `DTR`, `RTS`, `CTS`	Data Carrier Detect, Data Set Ready, Data Terminal Ready, Request To Send and Clear To Send. The serial port's control signals.
DIL	Digital Impairment Learning. The `V.90` and `V.92` sequence that measures the digital network.
`DIS`, `DTC`, `DCS`	Digital Identification Signal, Digital Transmit Command and Digital Command Signal. The fax messages that offer (`DIS`), ask for polling (`DTC`) and choose (`DCS`) the session's settings.
DLE	Data Link Escape (10h). In fax and voice modes, the character that starts a code in the data, such as `<DLE><ETX>`.
`DMT`	Discrete Multitone. Many carriers side by side, as in `V.wb` and `ADSL`.
DSVD	Digital Simultaneous Voice and Data. A conversation and a data connection on one modem call (`V.70`, `+DSVD`).
ECM	Error Correction Mode. Fax pages sent in frames, and repeated where damaged.
`EDT`	European Deaf Telephone. A text telephone at `110` bit/s (`V.18`).
`ETSI`	European Telecommunications Standards Institute. The European standards body.
`FSK`, `DPSK`, `QAM`	Frequency Shift Keying, Differential Phase Shift Keying and Quadrature Amplitude Modulation. Ways of putting data on a carrier: by its frequency, by its phase, by phase and size together.
`HDLC`	High-level Data Link Control. Frames with flags and an `FCS`, the basis of `LAPM`, of fax's `T.30` messages and of `PPP`.
`ISDN`	Integrated Services Digital Network. Digital telephone lines with 64000 bit/s channels.
ITU-T	International Telecommunication Union, Telecommunication Standardization Sector. The standards body whose recommendations are V.… (modems), T.… (fax) and so on.
Leased line	A permanent line between two modems, with no dialing; 2-wire or 4-wire (`&L`).
LR, LD	Link Request and Link Disconnect. `MNP`'s messages that start and end its link.
`MDMF`, `SDMF`	Multiple Data Message Format and Single Data Message Format. The two forms of a caller ID message.
MH	Modified Huffman. The plainest fax page coding (`T.4`).

Term	Meaning
Modem-on-hold	`V.92`'s way to put a data call on hold to answer another call (`+PMH`).
`PBX`	Private Branch Exchange. An office telephone system, such as Asterisk or FreeSWITCH.
`PPP`	Point-to-Point Protocol. How a computer carries the internet over a modem call.
`PREP`, `ACK`, `TERM`	Preparatory, Acknowledgement and Termination. `V.54`'s signals to start, confirm and end a remote loop test.
`PSTN`	Public Switched Telephone Network. The ordinary telephone network.
Retrain, renegotiation	Modems re-measuring the line during a call (retrain), or changing only the rate (renegotiation).
Robbed-bit signalling	A digital network taking a bit of some samples for its own use, which a `V.90` modem must allow for.
`RTTY`	Radio Teletypewriter. Baudot over radio, as radio amateurs send it.
`SNR`	Signal-to-Noise Ratio. How far the signal stands above the noise, in dB.
`SYN`	Synchronous idle. The idle octet of `V.80`'s synchronous mode.
T.30	The fax protocol: its messages, phases and timers.
Type II caller ID	The caller ID of a waiting call, sent during a call.
Unimodem	Windows' modem driver; `#UD` gives its diagnostics.
V.8	The start of a modern call, where the modems tell each other what they have.
`V.cvm`, `V.wb`	D-Modem's data over mobile speech paths, and its wideband modem for a dry pair.
X.75, V.110, V.120	Protocols on an `ISDN` data channel (`+B2`).
XID	Exchange Identification. `V.42`'s message that agrees the compression.
`XON`, `XOFF`	Transmit On and Transmit Off. The characters (DC1 and DC3) that start and stop the flow of data.
zstd	Zstandard. A modern compression D-Modem offers between two D-Modems (`+DSZ`).

Notes

21 Compatibility

21.1 Where D-Modem Differs

- `&K` is the compression, as on USRobotics' modems; on Rockwell's and Conexant's it is flow control. `&K3` leaves `V.42`'s compression on (`MNP 5` off), `&K0` turns it off.
- The S-registers past S16 are D-Modem's own: S29 and S30 are the speaker's (elsewhere the hook flash time and the inactivity timer), S92 the answer's delay (elsewhere a fax transmit level). The others (`S36`, `S37`, `S91` ...) are free: kept as written. S7 and S11 start at 60 and 100.
- There are no stored numbers or profiles: `&Z` and `ATDS=`*n* are `ERROR`, `&W` stores nothing.
- `ATI`'s numbers are the modem's own (`I0` to `I7`), but for the call statistics: `I6` as USRobotics', `I11` as Conexant's, `&V1` as Rockwell's.
- The numeric results (`V0`) and `CONNECT`'s number for each rate are D-Modem's own; those of USRobotics' and Conexant's modems differ.
- `MNP` is classes 2 to 5 (`\N`, `%C`): there is no `MNP 10` or 10EC. A modem with `MNP 10EC` connects with `V.42` (its "EM" detection patterns are passed over). See `docs/link.md`, `MNP`.
- K56flex is not there: `+MS=56` runs `V.90`.

21.2 Not Supported

These are `ERROR`: `&V2` and up, `&Z` (no stored numbers), `&T2` and `&T9` and up, `\B` (a break), `X5` and up, `ATI8` to `ATI10` and `ATI12` and up, `-V90` (`+MS` chooses), `+MA`, `+MSC`, `+EB`, `+ESR`, `+ETBM`, `+EWIND`, `+EFRAM`, `+A8M`, `+A8T`, `+PMHF`, `+GSN` and `+GOI`; `+MS`'s names of modulations the modem has not got (`V23S`, `V26B`, `V26T`, `V27` ...).

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22 Glossary

Term	Meaning
A-law, u-law	The two ways `G.711` codes telephone audio in 8 bits: A-law in Europe, u-law in North America and Japan.
`ADPCM`	Adaptive Differential Pulse Code Modulation. Compressed audio, one of the voice formats of `+VSM`.
`ADSI`	Analog Display Services Interface. A telephone's screen and keys, driven by tones.
`ALSA`	Advanced Linux Sound Architecture. The sound system of Linux; `dmodemd` uses it to reach a telephone line through a sound card.
`ASVD`	Analogue Simultaneous Voice and Data. Speech and data on one modem carrier (`V.61`, `+ASVD`).
Automode	Trying modulations from the fastest down until both modems have one (`N`, `+MS`).
Baudot	The 5-bit code (ITA2) of teleprinters, of the deaf community's `TTY`, of telex and of `RTTY`.
Bell 103, Bell 212A	The North American 300 and 1200 bit/s modulations, akin to `V.21` and `V.22`.
`CAPI`	Common `ISDN` Application Programming Interface. The programming interface of `ISDN` cards; `+B2` uses its numbers.
`CED`, `CNG`	Called terminal identification and Calling tone. The fax answer tone (2100 Hz) and the calling fax's tone (1100 Hz).
`CIG`, `CSI`, `TSI`	Calling subscriber identification, Called Subscriber Identification and Transmitting Subscriber Identification. The fax identities of the poller, the answerer and the caller (`+FLI`).
`Class 1`, `Class 2`, `Class 2.0`	The fax command sets: in `Class 1` (`T.31`) the `DTE` runs the fax protocol, in `Class 2` and `Class 2.0` (`T.32`) the modem does.
Clear channel	A 64000 bit/s digital channel used as it is, with no modulation, as in an `ISDN` data call.
`CPE`	Customer Premises Equipment. The equipment at the subscriber's end of a line.
`CRC`, `FCS`	Cyclic Redundancy Check and Frame Check Sequence. A checksum on a frame that shows whether it arrived damaged.

Notes

Notes